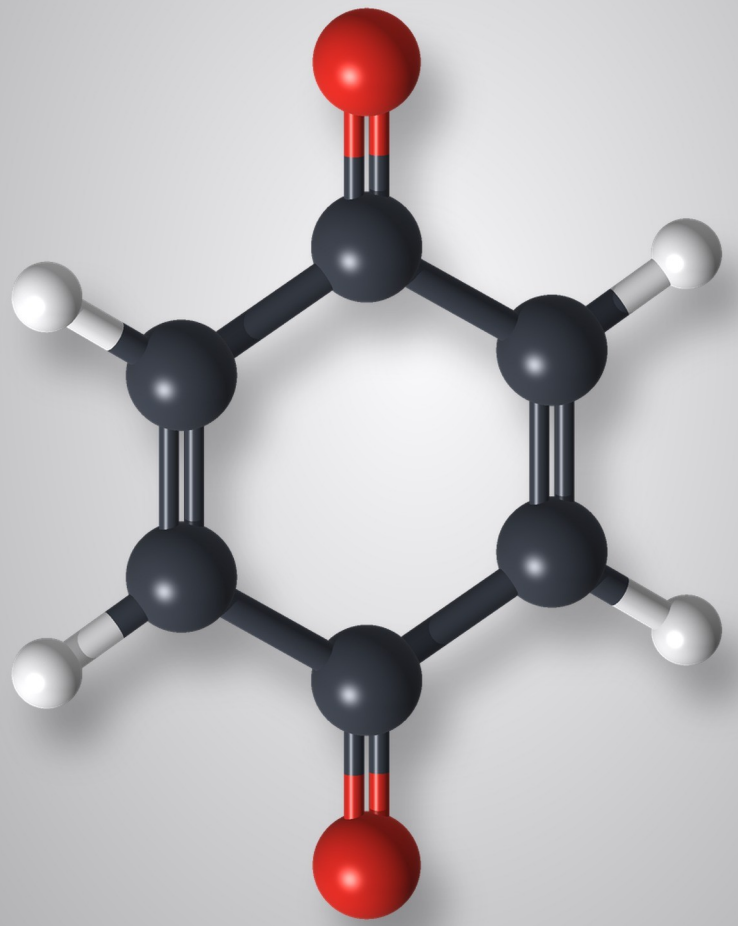
Introduction: redox chemistry, DFT and the Schrödinger suite

Session 1 of 5 · Molecular modelling and DFT with the Schrödinger suite

Nabil Khossossi | Autonomous Energy Materials Discovery, DIFFER

DIFFER

NWO



Session plan (2 hours)

25 min	30 min	24 min	9 min	20 min	11 min
0:00	0:25	0:55	1:19	1:28	1:48
The chemistry	**DFT background**	**Hands-on: Maestro**	**Command line**	**Hands-on: Epik**	**Wrap-up**
states of benzoquinone, the Nernst equation, the proton-electron grid, microstates	theory, units, frequencies, solvation, Gibbs energy, pK_a and potentials	project, structures, LigPrep, Jaguar panel	the same steps, typed	what it lists, how to enumerate	rules, homework, Session 2

Bring to the session

Your workstation with Maestro 2026-2 open · the schrodinger-tutorials-redox repository, folder 01-introduction · worksheet.md, printed or open in an editor



Place in the series and learning goals

S1 today	**Introduction**
S2	Optimisation and frequencies
S3	pK_a values and reduction potentials
S4	Automation with the Python API
S5	Microstates and E-pH diagrams

By the end of today you can

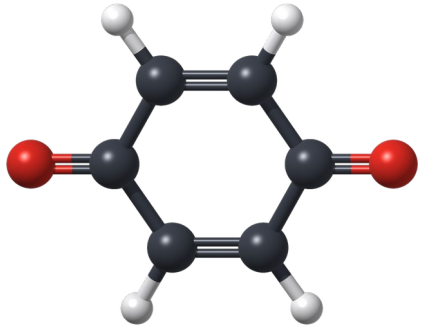
1. Name the benzoquinone states used here and explain the charge and spin of each
2. Explain what DFT computes and approximates, and what an optimisation, a frequency job and implicit water add
3. Find any panel in Maestro and choose where a job runs: localhost for LigPrep and Epik, a compute host for Jaguar
4. Prepare the states in Maestro and from the command line without changing their charge
5. Run Epik so that it lists the relevant protonation states, and say why it cannot replace Jaguar



Part 0 · The chemistry

Why one molecule needs many calculations

Quinone redox chemistry in an aqueous flow battery

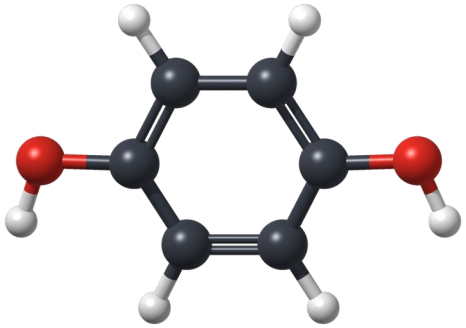


Q benzoquinone, oxidised

$+ 2\,e^- + 2\,H^+$



reduction / oxidation



QH_2 hydroquinone, reduced

In an aqueous flow battery

the molecule is dissolved in water and cycles between Q and QH_2 thousands of times.

The potential

E = −ΔG/zF; the cell voltage is the difference between the potentials of the two electrolytes

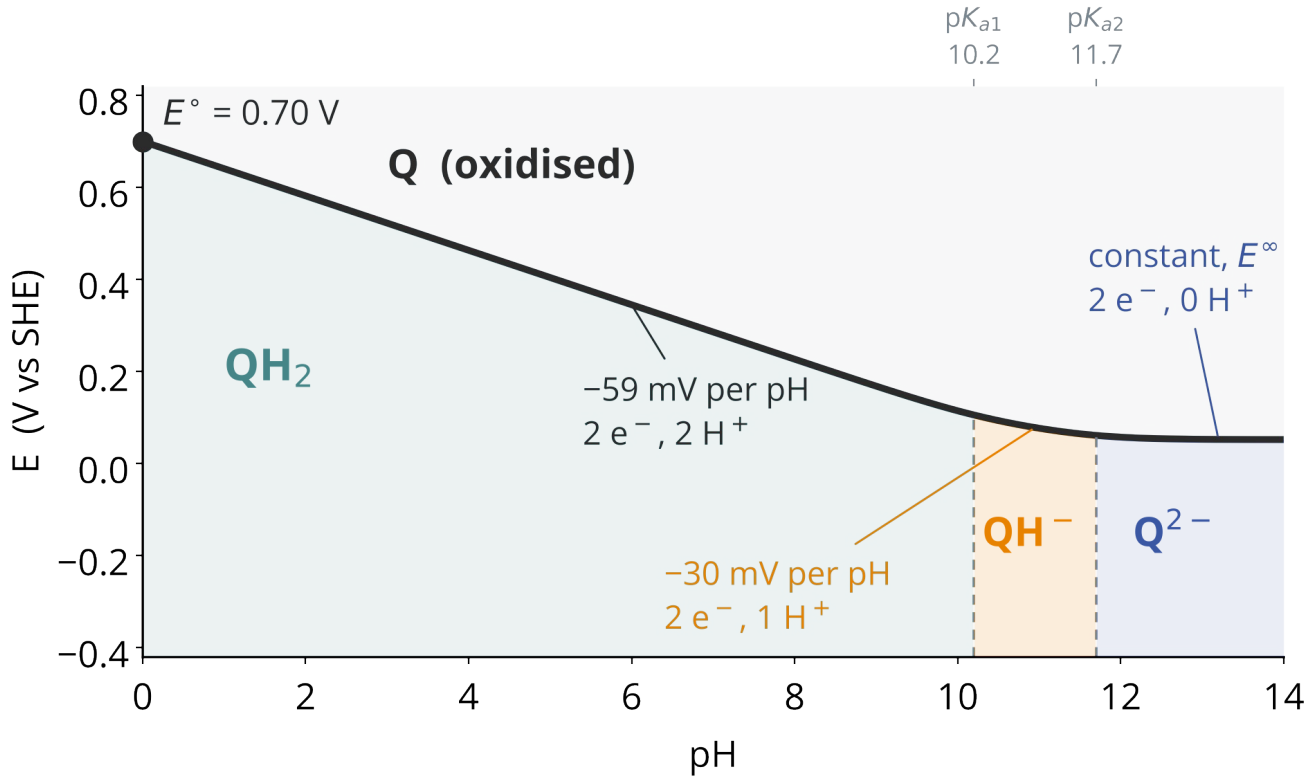
The protons

how many H^+ travel with the electrons depends on pH

The stability

a radical or anion that decomposes costs capacity

pH dependence of the reduction potential



E° = 0.70 V vs SHE, pK_{a1} = 10.2 and pK_{a2} = 11.7 of QH_2, as compiled by Gaudin and Aubry (2022); other compilations give 9.9 and 11.4. Line drawn for equal total concentrations of oxidised and reduced forms.

Below pH 10.2

two electrons and two protons: the potential falls 59 mV per pH unit

Between pK_{a1} and pK_{a2}

two electrons, one proton: the potential falls 30 mV per pH unit

Above pH 11.7

two electrons, no protons: the potential is constant, E^{∞} of the couple Q/Q^{2-}

Aim of the series: compute this line from DFT Gibbs energies, and test it against experiment.

The Nernst equation and the reference electrode

A potential is written for the **reduction** Ox + z e⁻ → Red (IUPAC). At activities a:

$$E = E^\circ - \frac{RT}{zF} \ln \frac{a_{\mathrm{Red}}}{a_{\mathrm{Ox}}}$$

When m protons travel with the z electrons, each pH unit shifts the potential by

$$\mathrm{Ox} + z\,e^- + m\,\mathrm{H^+} \rightleftharpoons \mathrm{Red}: \qquad \frac{\partial E}{\partial\,\mathrm{pH}} = -\frac{m}{z}\,\frac{RT\ln 10}{F} = -\frac{m}{z} \times 59.2\ \mathrm{mV}$$

For Q/QH_2 (z = 2), with the reduced form a diprotic acid, this gives the line of the previous slide:

$$E(\mathrm{pH}) = E^\circ - k\,\mathrm{pH} + \frac{k}{2}\log_{10}\left(1 + 10^{\,\mathrm{pH}-\mathrm{p}K_{\mathrm{a1}}} + 10^{\,2\mathrm{pH}-\mathrm{p}K_{\mathrm{a1}}-\mathrm{p}K_{\mathrm{a2}}}\right), \qquad k = \frac{RT\ln 10}{F}$$

E°: all activities 1, so pH 0. $E^{\circ\prime}$: formal potential at a stated pH and medium. $E_{1/2}$ from cyclic voltammetry approximates $E^{\circ\prime}$.

Reference electrodes, 25 °C

electrode	E vs SHE
SHE	0 V by definition
Ag/AgCl, sat. KCl	+0.197 V
Ag/AgCl, 3 M KCl	+0.210 V
SCE, sat. KCl	+0.241 V
RHE	−0.0592 V × pH

All potentials in this course are reduction potentials against the SHE. On the RHE scale, the 2 e⁻/2 H⁺ segment of the line is flat.

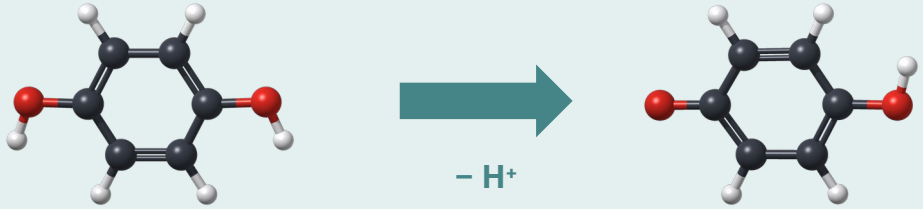


IUPAC Green Book, 3rd ed. (2007). Reference electrode values: A. J. Bard and L. R. Faulkner, Electrochemical Methods, 2nd ed. (2001).

Two coordinates: protons and electrons

COORDINATE 1 · PROTONS, n_H

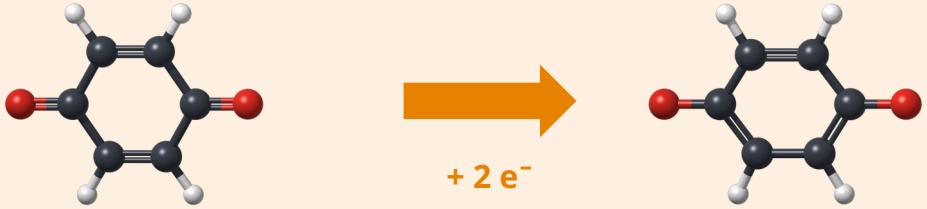
Acid-base chemistry



Same electrons, one proton less. Gives pK_a and which form dominates at each pH.

COORDINATE 2 · ELECTRONS, n_e

Redox chemistry



Same protons, two electrons more. Gives reduction potentials.

E(pH) needs both coordinates. Every pK_a and potential in this course is a difference of Gibbs energies G taken from one consistent table.

Charge and spin multiplicity from the two counts

RULE 1 · charge

$$q = q_{\text{parent}} + n_{\text{H}} - n_e$$

RULE 2 · spin multiplicity

$$2S+1 = \begin{cases} 2, & n_e \text{ odd} \\ 1, & n_e \text{ even} \end{cases}$$

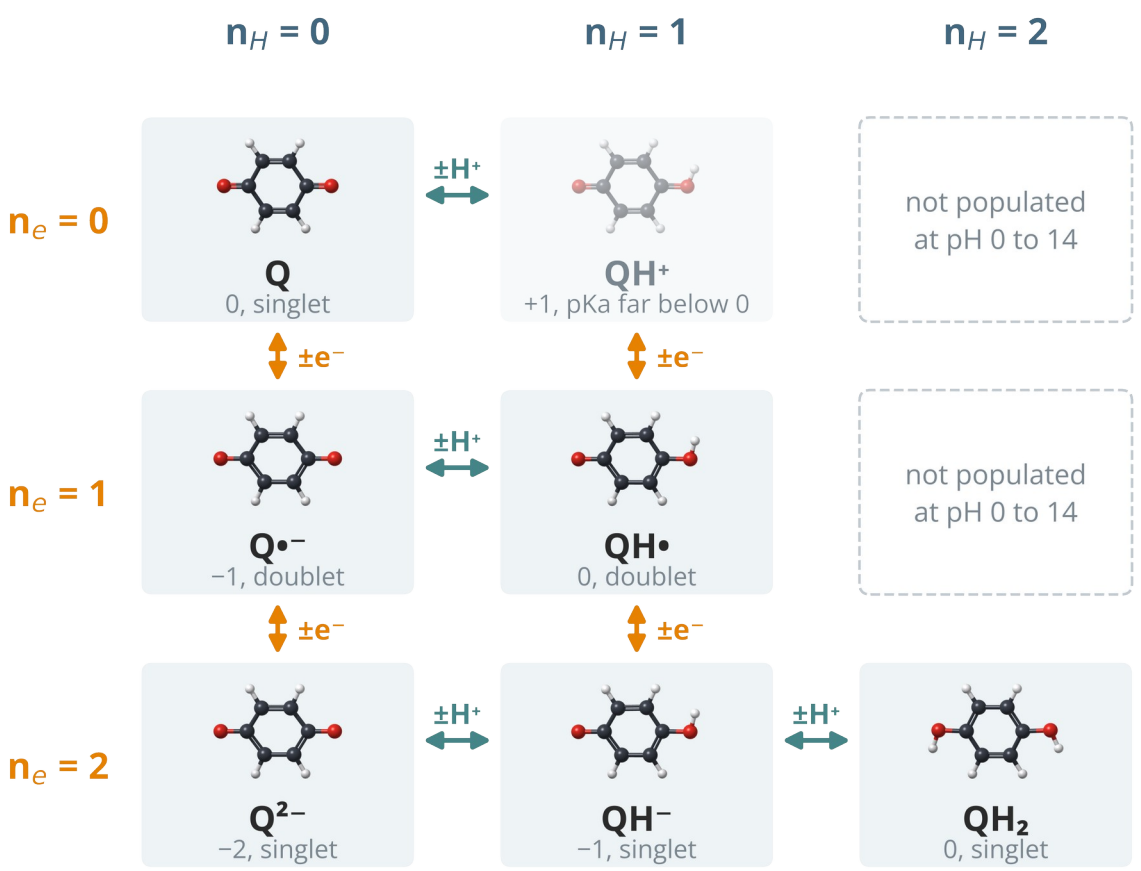
state	charge	n_e	multiplicity
QH^-	1 − 2 = −1	2, even	1, singlet
QH•	1 − 1 = 0	1, odd	2, doublet (radical)
Q^{2-}	0 − 2 = −2	2, even	1, singlet

Jaguar needs both

They are the keywords molchg and multip. A wrong pair stops the job or converges to the wrong state.

Rule 2 assumes a closed-shell parent and the low-spin state. Run doublets spin-unrestricted (Jaguar: iuhf=1; its default is restricted) and check spin contamination: $\langle S^2 \rangle$ about 0.75. For dianions, also check the triplet. For a charged parent such as MV^{2+}, q_parent ≠ 0.

States of benzoquinone on a proton-electron grid



Horizontal: proton transfer, at constant n_e

$$\mathrm{HA} \rightleftharpoons \mathrm{A^-} + \mathrm{H^+}, \qquad \Delta G^\circ = RT \ln 10 \; \mathrm{p}K_\mathrm{a}$$

Epik predicts micro-pK_a values site by site and lists the protonation states and tautomers present in a pH window. It moves along a row in both directions: adding H^+ to a basic site or removing it from an acidic one.

Vertical: electron transfer, at constant n_H

$$\mathrm{Ox} + z\,e^- \rightleftharpoons \mathrm{Red}, \qquad \Delta G^\circ = -zF\,E^\circ$$

The electron comes from an electrode or a reductant; this is not an acid-base equilibrium. Epik keeps the electron count of its input, so it never leaves the row it starts in.

Consequence: one input structure per row (n_e = 0, 1, 2), then Epik on each closed-shell row. The overall reaction Q + 2e⁻ + 2H⁺ → QH_2 is the diagonal; which cells are populated at a given pH sets E(pH).

Proton-coupled electron transfer thermochemistry: J. J. Warren, T. A. Tronic and J. M. Mayer, Chem. Rev. 110, 6961 (2010). Epik: R. C. Johnston et al., J. Chem. Theory Comput. 19, 2380 (2023). Molecules: force-field geometries.

Microstates, macrostates and enumeration

Microstate: one exact structure: proton positions, charge and electron count fixed. Conformers (OH rotamers) belong to it: use the lowest-G one, or Boltzmann-sum near-degenerate ones.

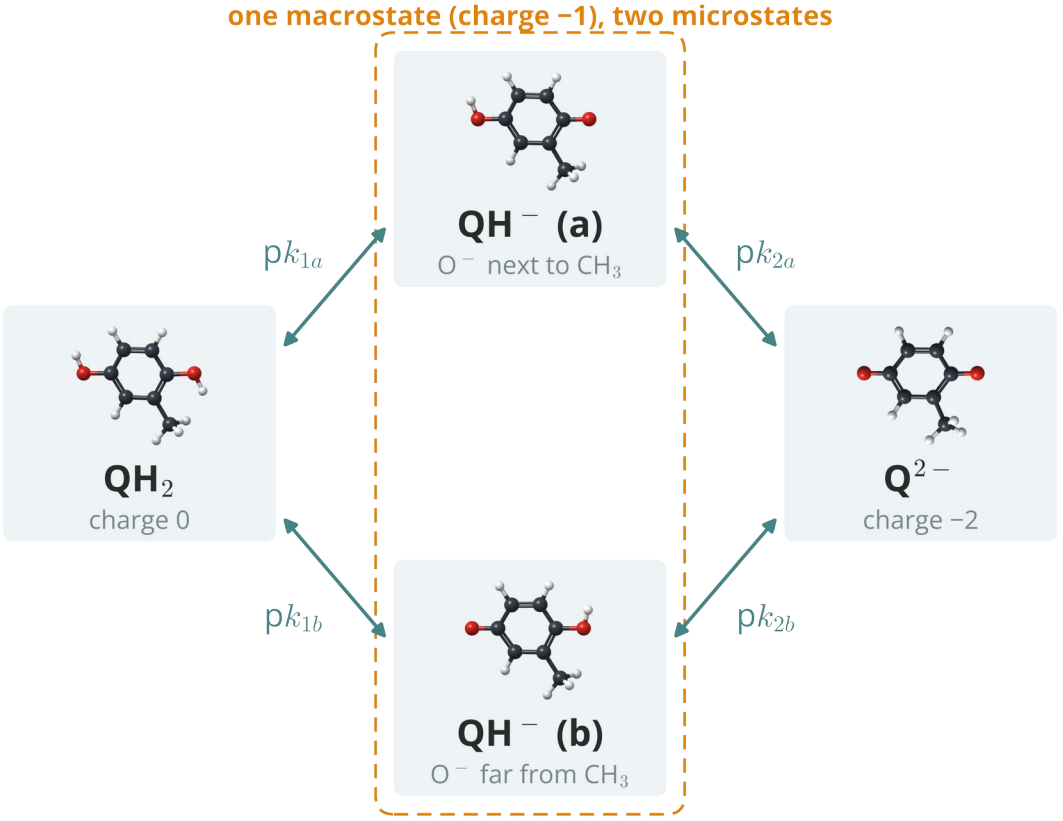
Macrostate: all microstates with the same n_H and n_e, one cell of the proton-electron grid. Experiments see macrostates.

Tautomers: microstates of one cell that differ only in where a proton sits.

Micro- and macro-pK_a: between two microstates (pk), or between two macrostates (pK).

Enumeration: generating every relevant microstate of every cell. Rows (n_e) are built by us; protonation states and tautomers within a row come from Epik.

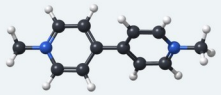
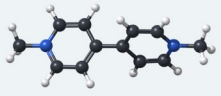
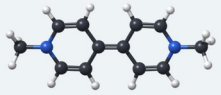
$$K_1 = k_{1a} + k_{1b}, \qquad \frac{1}{K_2} = \frac{1}{k_{2a}} + \frac{1}{k_{2b}}, \qquad k_{1a}k_{2a} = k_{1b}k_{2b} = K_1K_2$$



2-Methylhydroquinone: the two OH groups are no longer equivalent, so the monoanion cell holds two microstates.

Hydroquinone: QH^- has two identical microstates (statistical factor log 2). It enters through the symmetry numbers in S or by counting by hand, never both. Molecules: force-field geometries.

Another molecule: methyl viologen moves only vertically

	$n_H = 0$	$n_H = 1$	$n_H = 2$
$n_e = 0$	 **MV^{2+}** +2, singlet	no acidic or basic site	no acidic or basic site
	±e⁻		
$n_e = 1$	 **$MV^{\bullet+}$** +1, doublet	no acidic or basic site	no acidic or basic site
	±e⁻		
$n_e = 2$	 **MV^{0}** 0, singlet	no acidic or basic site	no acidic or basic site

Methyl viologen (1,1'-dimethyl-4,4'-bipyridinium, MV^{2+}) is used as a negolyte (the electrolyte on the negative side) in pH-neutral aqueous organic flow batteries. Both nitrogens carry a methyl group, so the molecule has no acidic or basic site.

Its one-electron reductions, $MV^{2+} \rightarrow MV^{\bullet+}$ ($E^\circ \approx -0.45$ V vs SHE) $\rightarrow MV^0$, involve no protons, so the potentials do not depend on pH. MV^0 is poorly soluble; flow cells use only the first electron.

What Epik gives: run on any of the three structures, it should return that structure only. **What it does not give:** the other two rows. They exist only if we build them.

Compare benzoquinone: three rows with up to three protonation states each, and pH-dependent potentials. The proton-electron grid tells you which calculations a molecule needs.

T. Liu et al., Adv. Energy Mater. 6, 1501449 (2016). Molecules: force-field geometries; $MV^{\bullet+}$ is drawn as one resonance structure, the unpaired electron is delocalised over both rings.

The benzoquinone states used in this tutorial

key	label	n_H	n_e	charge	mult	benzoquinone SMILES	description
Q	**Q**	0	0	0	1	O=C1C=CC(=O)C=C1	quinone, oxidised
Qradm	**Q•⁻**	0	1	−1	2	[O-]c1ccc([O])cc1	semiquinone radical anion
QHrad	**QH•**	1	1	0	2	[O]c1ccc(O)cc1	neutral semiquinone radical
QH2	**QH_2**	2	2	0	1	Oc1ccc(O)cc1	hydroquinone, reduced
QHm	**QH^-**	1	2	−1	1	[O-]c1ccc(O)cc1	monoanion
Q2m	**Q^{2-}**	0	2	−2	1	[O-]c1ccc([O-])cc1	dianion

n_H and n_e are counted from the neutral oxidised parent Q. Keys name the files (bq_QH2, bq_QHm). QH• and Q•⁻ are open-shell doublets. QH^+ (protonated carbonyl, pK_a far below 0) is left out; protonated cations matter for N-heterocycles such as phenazine.



Part 1 · DFT background

What DFT is, how a calculation works, and what it gives: energy, geometry, frequencies, solvation and Gibbs energy

What a DFT calculation takes and gives

1. **The input**

   positions of the nuclei, total charge and spin (molchg, multip)

2. **The approximations**

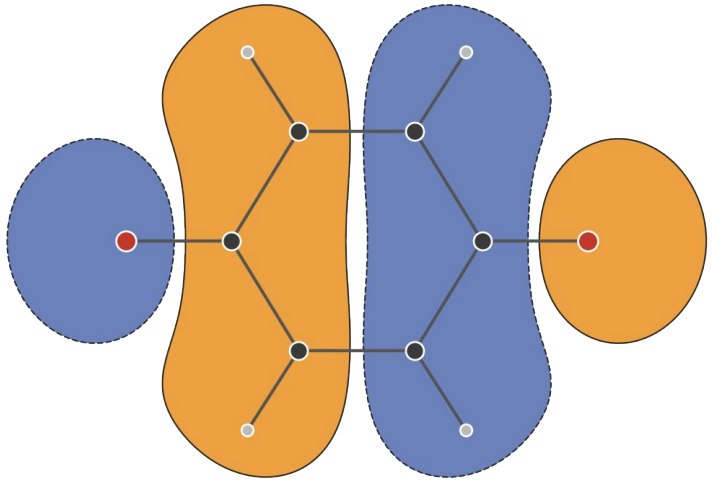
   a functional for exchange and correlation (B3LYP) and a basis set for the orbitals (def2-TZVPD); both are explained in the next slides

3. **The output**

   the electronic energy E of that geometry, the electron density and the orbitals

4. **The cost**

   grows steeply with the number of basis functions: diffuse functions on anions are expensive but needed



LUMO of benzoquinone. Figures use B3LYP/def2-SVP for speed; the course uses def2-TZVPD.

Units and conventions in this course

quantity	value
1 hartree (E_h), Jaguar's energy unit	627.51 kcal/mol = 27.211 eV = 2625.5 kJ/mol
1 eV	23.06 kcal/mol
1 bohr	0.52918 Å
atomic units	$\hbar = m_e = e = 4\pi\varepsilon_0 = 1$
RT at 298.15 K	0.592 kcal/mol
RT ln 10	1.364 kcal/mol, 1 pK_a unit
RT ln 10 / F	59.16 mV per pH unit for 1 e^- and 1 H^+
F	23.06 kcal/(mol·V)

Conventions

Jaguar prints energies in hartree. This course reports Gibbs energies in kcal/mol, potentials in V vs SHE.

Temperature 298.15 K throughout.

Gibbs energies of solutes at 1 mol/L (next slides).



The molecular Schrödinger equation

For N electrons with space-spin coordinates $\mathbf{x}_i$ and M nuclei with positions $\mathbf{R}_A$, charges Z_A and masses M_A, in atomic units:

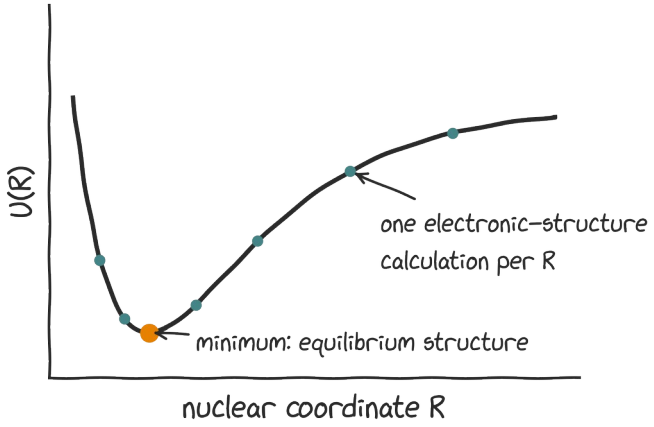
$$\hat{H}\,\Psi(\mathbf{x}_1,\ldots,\mathbf{x}_N;\mathbf{R}_1,\ldots,\mathbf{R}_M) = E\,\Psi(\mathbf{x}_1,\ldots,\mathbf{x}_N;\mathbf{R}_1,\ldots,\mathbf{R}_M) \tag{1}$$

$$\hat{H} = \underbrace{-\sum_{i=1}^{N}\tfrac{1}{2}\nabla_i^2}_{\hat{T}_\mathrm{e}} \underbrace{-\sum_{A=1}^{M}\frac{1}{2M_A}\nabla_A^2}_{\hat{T}_\mathrm{N}} \underbrace{-\sum_{i,A}\frac{Z_A}{|\mathbf{r}_i-\mathbf{R}_A|}}_{\hat{V}_\mathrm{eN}} + \underbrace{\sum_{i<j}\frac{1}{|\mathbf{r}_i-\mathbf{r}_j|}}_{\hat{V}_\mathrm{ee}} + \underbrace{\sum_{A<B}\frac{Z_A Z_B}{|\mathbf{R}_A-\mathbf{R}_B|}}_{V_\mathrm{NN}} \tag{2}$$

Born-Oppenheimer approximation. Since $m_e/M_A < 1/1836$, the electrons are solved for fixed nuclei:

$$\hat{H}_\mathrm{el}\,\Psi_\mathrm{el}(\mathbf{x}_1,\ldots,\mathbf{x}_N;\mathbf{R}) = E_\mathrm{el}(\mathbf{R})\,\Psi_\mathrm{el}(\mathbf{x}_1,\ldots,\mathbf{x}_N;\mathbf{R}), \qquad U(\mathbf{R}) = E_\mathrm{el}(\mathbf{R}) + V_\mathrm{NN}(\mathbf{R}) \tag{3}$$

$U(\mathbf{R})$ is the potential energy surface: its minima are equilibrium structures, its curvature gives the vibrational frequencies.



1,4-Benzoquinone, $C_6H_4O_2$: $N = 56$, so Ψ depends on $3N = 168$ spatial coordinates and 56 spins. Because V_ee couples all electrons, no closed-form solution exists beyond one-electron systems.

M. Born and R. Oppenheimer, Ann. Phys. 389, 457 (1927).

The electron density and the Hohenberg-Kohn theorems

The ground-state density is a function of three variables only:

$$n(\mathbf{r}) = N \int |\Psi(\mathbf{r}s_1, \mathbf{x}_2, \ldots, \mathbf{x}_N)|^2 \, ds_1 \, d\mathbf{x}_2 \cdots d\mathbf{x}_N, \qquad \int n(\mathbf{r}) \, d\mathbf{r} = N \tag{4}$$

Theorem 1. The ground-state density fixes v_{ext} up to a constant:

$$v_{\text{ext}}(\mathbf{r}) = -\sum_A \frac{Z_A}{|\mathbf{r} - \mathbf{R}_A|} \quad \longleftrightarrow \quad n_0(\mathbf{r}) \tag{5}$$

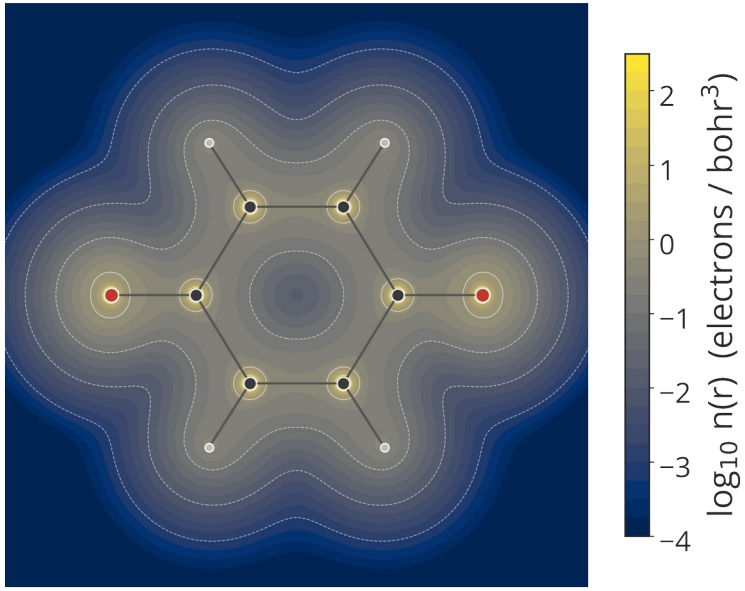
So every ground-state property is a **functional** of n_0: a rule that maps a whole function to a number,

$$\text{function } n(\mathbf{r}) \longmapsto \text{number } E[n], \qquad \text{e.g. } E_{\text{ext}}[n] = \int v_{\text{ext}}(\mathbf{r}) \, n(\mathbf{r}) \, d\mathbf{r}$$

Theorem 2. The energy functional is minimised by the true ground-state density:

$$E_v[n] = F_{\text{HK}}[n] + \int v_{\text{ext}}(\mathbf{r}) \, n(\mathbf{r}) \, d\mathbf{r} \;\geq\; E_0, \qquad F_{\text{HK}}[n] = \langle \Psi[n] | \hat{T}_e + \hat{V}_{ee} | \Psi[n] \rangle \tag{6}$$

$F_{\text{HK}}[n]$ is universal, the same for every molecule, but its explicit form is unknown. A formal definition is in the appendix.



n(r) of 1,4-benzoquinone in the ring plane, logarithmic scale. B3LYP/def2-SVP, computed with PySCF.

P. Hohenberg and W. Kohn, Phys. Rev. 136, B864 (1964); M. Levy, Proc. Natl. Acad. Sci. USA 76, 6062 (1979).

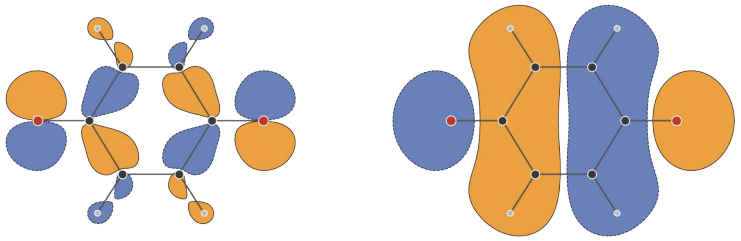
The Kohn-Sham scheme

A reference system of N non-interacting electrons in orbitals φ_i with the same density as the real system:

$$n(\mathbf{r}) = \sum_{i=1}^{N} |\varphi_i(\mathbf{r})|^2 \qquad (7)$$

$$E[n] = \underbrace{-\tfrac{1}{2}\sum_i \langle\varphi_i|\nabla^2|\varphi_i\rangle}_{T_s[n]} + \int v_{\text{ext}}\, n\, d\mathbf{r} + \underbrace{\frac{1}{2}\iint \frac{n(\mathbf{r})\, n(\mathbf{r}')}{|\mathbf{r}-\mathbf{r}'|}\, d\mathbf{r}\, d\mathbf{r}'}_{J[n]} + E_{\text{xc}}[n] \qquad (8)$$

$E_{\text{xc}}[n] \equiv \big(T[n] - T_s[n]\big) + \big(V_{\text{ee}}[n] - J[n]\big)$ the only term that is approximated.



Kohn-Sham HOMO (O lone pairs, ring plane) and LUMO (π, 1 Å above the plane) of 1,4-benzoquinone; B3LYP/def2-SVP.*

Minimising (8) with respect to orthonormal orbitals gives the Kohn-Sham equations:

$$\Big[-\tfrac{1}{2}\nabla^2 + v_{\text{ext}}(\mathbf{r}) + v_{\text{H}}(\mathbf{r}) + v_{\text{xc}}(\mathbf{r})\Big]\varphi_i(\mathbf{r}) = \varepsilon_i\, \varphi_i(\mathbf{r}), \qquad v_{\text{H}} = \int \frac{n(\mathbf{r}')}{|\mathbf{r}-\mathbf{r}'|} d\mathbf{r}', \quad v_{\text{xc}} = \frac{\delta E_{\text{xc}}}{\delta n(\mathbf{r})} \qquad (9)$$

v_{H} and v_{xc} depend on n, and n on the orbitals: the equations are non-linear and are solved self-consistently. E_{xc} contains exchange (from antisymmetry; it also removes each electron's repulsion with itself) and correlation (electrons avoiding each other beyond the average field). The ε_i are not reduction potentials.



W. Kohn and L. J. Sham, Phys. Rev. 140, A1133 (1965).

Basis-set expansion and the self-consistent field

The orbitals are expanded in K atom-centred Gaussian basis functions χ_μ:

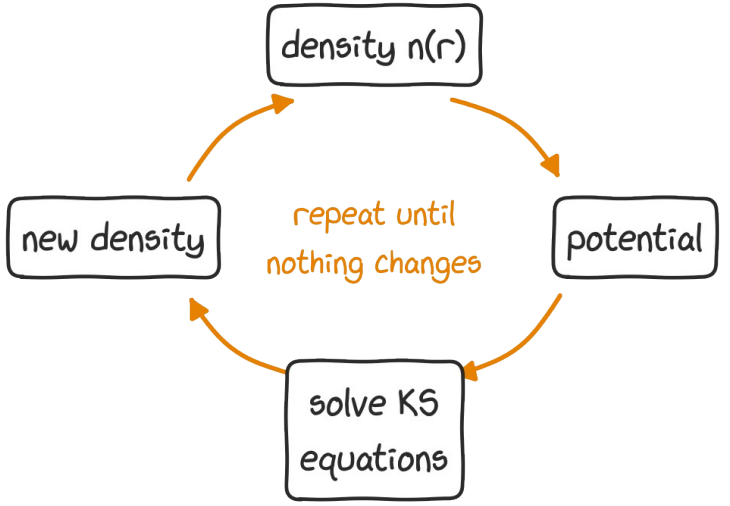
$$\varphi_i(\mathbf{r}) = \sum_{\mu=1}^{K} C_{\mu i}\,\chi_\mu(\mathbf{r}) \qquad (10)$$

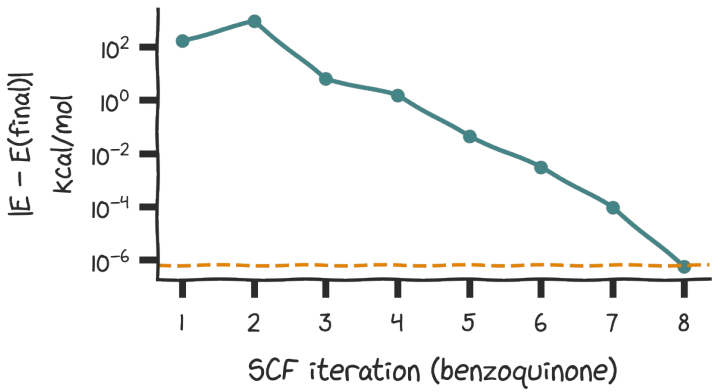
Each basis function is a fixed combination of Gaussians centred on atom A ($x_A = x - X_A$); def2 sets use their spherical combinations (5 d, 7 f), which gives the counts below:

$$\chi_\mu(\mathbf{r}) = \sum_k d_{k\mu}\, x_A^a\, y_A^b\, z_A^c\, e^{-\alpha_k |\mathbf{r}-\mathbf{R}_A|^2} \qquad \text{(contracted Gaussian on atom } A\text{)}$$

Inserting (10) into (9) gives a matrix equation for the coefficients C. The Kohn-Sham matrix depends on the density, which depends on the coefficients, so the equations are solved iteratively, the **self-consistent field (SCF)**: guess a density, build the matrix, solve, build a new density, repeat until the energy changes by less than a threshold. Matrix form: appendix, (A2) and (A3).

Benzoquinone: $K = 132$ (def2-SVP) and $K = 338$ (def2-TZVPD). The cost per iteration grows formally as K^3 to K^4. A geometry optimisation repeats the SCF at every point of $U(\mathbf{R})$; a frequency calculation adds the second derivatives.

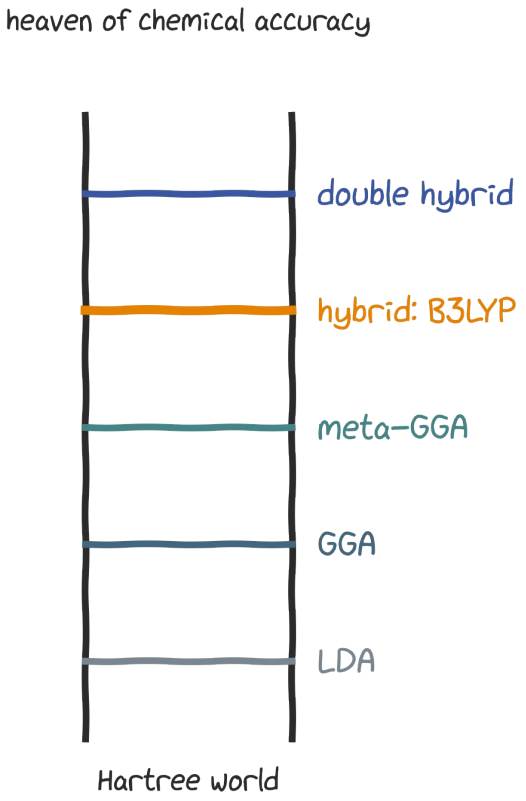




C. C. J. Roothaan, Rev. Mod. Phys. 23, 69 (1951); G. G. Hall, Proc. R. Soc. A 205, 541 (1951). Convergence data: B3LYP/def2-SVP, PySCF. The plot shows the distance to the converged energy; Jaguar prints the change between iterations.

Approximations to the exchange-correlation energy

LDA	$E_{\mathrm{xc}}^{\mathrm{LDA}} = \int n\, \epsilon_{\mathrm{xc}}^{\mathrm{unif}}(n)\, d\mathbf{r}$	*SVWN*
GGA	$E_{\mathrm{xc}}^{\mathrm{GGA}} = \int f(n, \nabla n)\, d\mathbf{r}$	*PBE, BLYP*
meta-GGA	$E_{\mathrm{xc}}^{\mathrm{mGGA}} = \int f(n, \nabla n, \tau)\, d\mathbf{r}, \quad \tau = \frac{1}{2}\sum_i \lvert\nabla\varphi_i\rvert^2$	*TPSS, SCAN*
hybrid	$E_{\mathrm{xc}}^{\mathrm{hyb}} = a\, E_{\mathrm{x}}^{\mathrm{HF}} + (1-a)\, E_{\mathrm{x}}^{\mathrm{GGA}} + E_{\mathrm{c}}^{\mathrm{GGA}}$	*B3LYP, PBE0*



B3LYP combines exact exchange with local and gradient-corrected terms:

$$E_{\mathrm{xc}}^{\mathrm{B3LYP}} = (1-a_0)E_{\mathrm{x}}^{\mathrm{LSDA}} + a_0 E_{\mathrm{x}}^{\mathrm{HF}} + a_{\mathrm{x}}\Delta E_{\mathrm{x}}^{\mathrm{B88}} + (1-a_{\mathrm{c}})E_{\mathrm{c}}^{\mathrm{VWN}} + a_{\mathrm{c}}E_{\mathrm{c}}^{\mathrm{LYP}}$$

with a_0 = 0.20, a_{x} = 0.72, a_{c} = 0.81, where $E_{\mathrm{x}}^{\mathrm{HF}}$ is exact (Hartree-Fock) exchange computed from the orbitals. B3LYP has no long-range dispersion; Grimme's D3 term (appendix) adds it. Its effect on these deprotonation and reduction energies is expected to be small; Session 2 tests it (dftname=B3LYP-D3).



Jacob's ladder after J. P. Perdew and K. Schmidt, AIP Conf. Proc. 577, 1 (2001); A. D. Becke, J. Chem. Phys. 98, 5648 (1993); P. J. Stephens et al., J. Phys. Chem. 98, 11623 (1994); S. Grimme et al., J. Chem. Phys. 132, 154104 (2010).

The functional and basis set used here

The functional: exchange and correlation

family	example	used for
GGA	PBE	cheap screening
hybrid GGA	**B3LYP**	redox and pK_a (this tutorial)
range-separated	ωB97X-D	charge transfer

A hybrid mixes in a fraction of exact exchange (20 % in B3LYP). Pure GGAs spread an extra electron too much, which matters for anions.

The basis set: how orbitals are shaped

def2- Karlsruhe family (Weigend and Ahlrichs 2005; diffuse sets: Rappoport and Furche 2010)

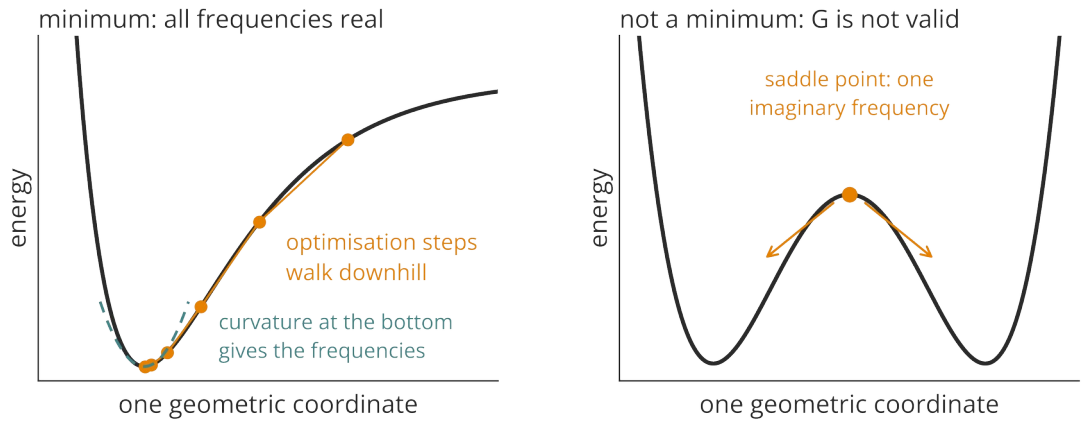
TZV triple-zeta valence: three functions per valence orbital

P polarisation functions: orbitals can deform

D diffuse augmentation of def2-TZVP: functions that extend far from the atoms

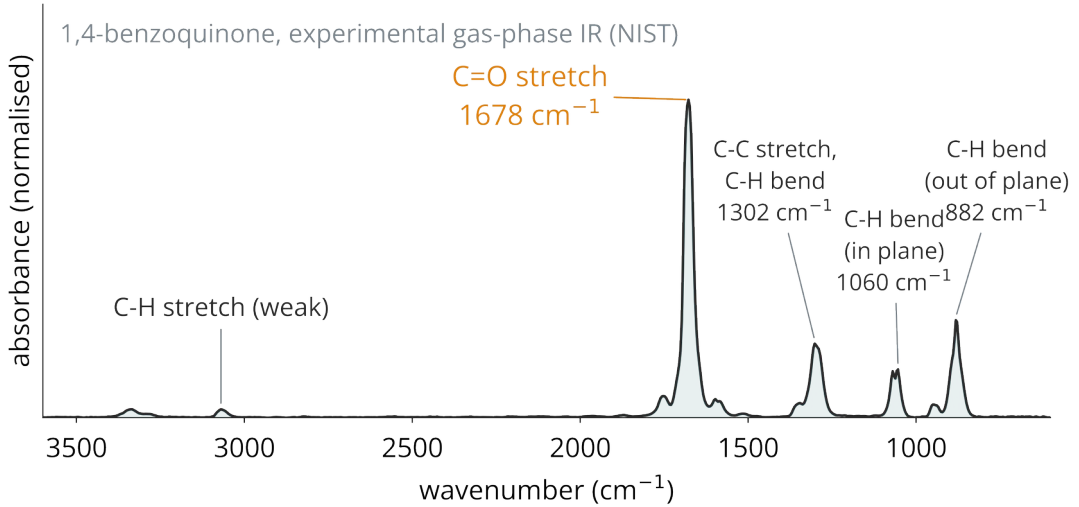
In this tutorial: B3LYP / def2-TZVPD in PBF water. Anions such as QH^- and Q^{2-} need the diffuse functions; without them the anion energy is too high and the computed pK_a comes out too high.

Geometry optimisation and vibrational frequencies



Model curves: a minimum (left) and a saddle point (right).

$$\tilde{\nu} = \frac{1}{2\pi c}\sqrt{\frac{k}{\mu}}, \qquad k = \frac{\partial^2 U}{\partial q^2} < 0 \;\Rightarrow\; \tilde{\nu} \text{ imaginary}$$



Experimental gas-phase IR of 1,4-benzoquinone (NIST). The strong bands are IR-active fundamentals; of the 30 vibrational modes, only the IR-active ones appear.

`igeopt=1`

move the atoms downhill until the forces vanish

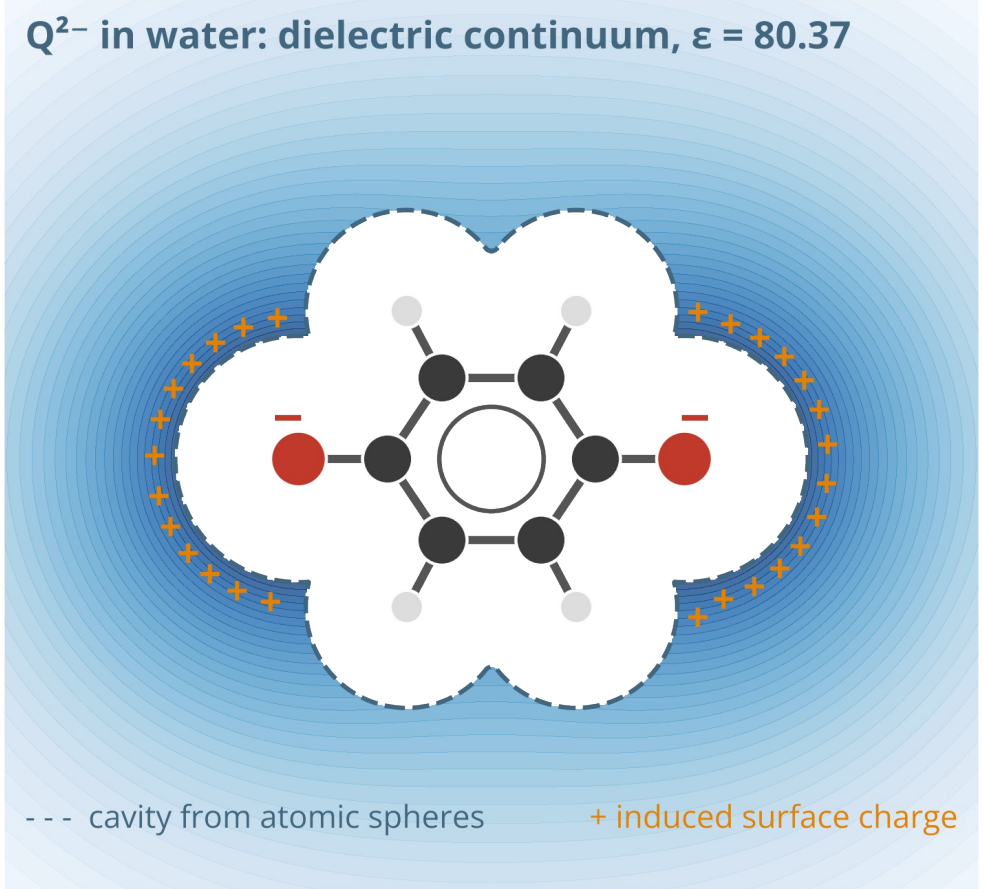
`ifreq=1`

curvature at the optimised geometry, same level of theory: all real means a true minimum

`imaginary mode`

printed as a negative wavenumber. One or more: a saddle point; distort along the mode and optimise again

Implicit solvation with PBF



Implicit solvent

water is a polarisable continuum around a cavity shaped like the molecule (Jaguar water: ε = 80.37, probe radius 1.40 Å)

Effect of water

charges are screened: a −2 anion may not be bound in vacuum, yet is stable in water. Missing: specific hydrogen bonds to anions

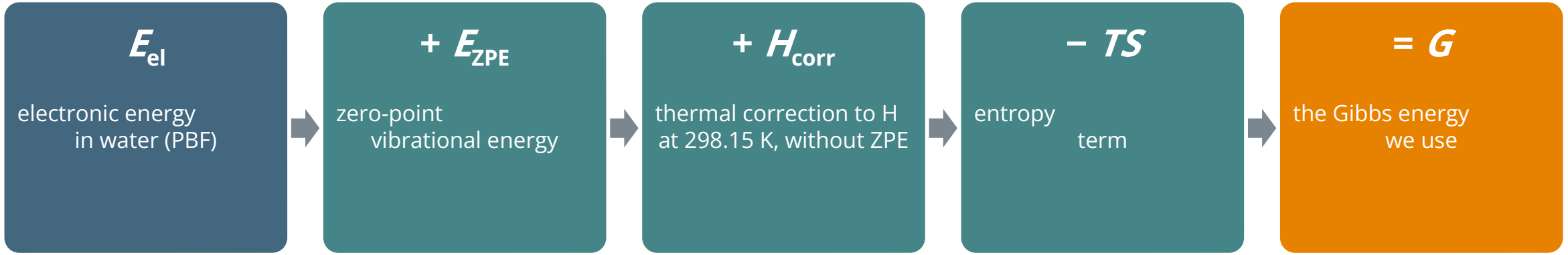
PBF in Jaguar

isolv=2: Poisson-Boltzmann finite-element solver, plus a cavity term and, in water, a first-shell correction

nogas=2

no gas-phase optimisation; optimisation and frequencies are done in water only

From electronic energy to Gibbs energy G



ZPE, H and S

from frequencies, masses and geometry: ideal gas, rigid rotor, harmonic oscillator at 1 atm. S uses the symmetry number σ

Standard state

add +1.89 kcal/mol to every solute to go from 1 atm to 1 mol/L (next slide)

In Jaguar's output

use Gtot, built on the solution-phase energy; it already contains the ZPE. A doublet has R ln 2 of electronic entropy: check the 'elec.' row

Standard states and reference values

Jaguar's thermal corrections are for an ideal gas at $p = 1$ atm. Solution chemistry uses $V^\circ = 1$ L per mol, so every solute gets

$$\Delta G^{\,1\,\mathrm{atm}\to 1\,\mathrm{M}} = RT\ln\frac{RT}{p\,V^\circ} = RT\ln 24.46 = +1.89\ \mathrm{kcal\,mol^{-1}}\quad \text{per species}$$

The Gibbs energy of the proton in water at 1 mol/L is built from experiment, not computed:

$$G^*(\mathrm{H^+},\mathrm{aq}) = \underbrace{-6.28}_{\text{gas, 1 atm}} + \underbrace{1.89}_{1\,\mathrm{atm}\to 1\,\mathrm{M}} \underbrace{-\,265.9}_{\Delta G^*_{\mathrm{solv}}(\mathrm{H^+}),\ 1\,\mathrm{M}\to 1\,\mathrm{M}} = -270.3\ \mathrm{kcal\,mol^{-1}}$$

The −265.9 kcal/mol is the 1 M → 1 M value (Kelly et al.); Tissandier's −264.0 already contains the 1 atm → 1 M step, so 1.89 is not added to it. −270.3 kcal/mol goes with the absolute SHE potential **4.28 V**: both treat the electron at rest (Fermi-Dirac convention). The IUPAC value 4.44 V includes the surface potential of water; mixing it in shifts every computed potential by 0.16 V.

Leaving out the 1.89 kcal/mol for the proton gives −272.2 kcal/mol, and every pK_a comes out 1.4 units too low.

For E° of Q/QH_2, $2G^*(H^+) - 2FE_{SHE}^{abs}$ equals the experimental G of H_2(g): the proton solvation energy drops out of a 2 e^-/2 H^+ couple.



M. D. Tissandier et al., J. Phys. Chem. A 102, 7787 (1998); C. P. Kelly, C. J. Cramer and D. G. Truhlar, J. Phys. Chem. B 110, 16066 (2006); A. A. Isse and A. Gennaro, J. Phys. Chem. B 114, 7894 (2010).

pK_a, E° and E^{∞} from the Gibbs energies

With one Gibbs energy G per state (1 mol/L), every observable is a difference of table entries:

Acidity, pK_a = −log K_a: one proton removed at constant electron count:

$$\mathrm{p}K_\mathrm{a} = \frac{G(\mathrm{A}^-) + G(\mathrm{H}^+) - G(\mathrm{HA})}{RT\ln 10} \tag{11}$$

Standard reduction potential $E°$: two electrons and two protons, at pH 0:

$$E^\circ = -\frac{G(\mathrm{QH_2}) - G(\mathrm{Q}) - 2\,G(\mathrm{H}^+)}{2F} - E^{\mathrm{abs}}_{\mathrm{SHE}} \tag{12}$$

High-pH potential E^{∞}, the couple Q/Q^{2-}: two electrons and no protons:

$$E^\infty = -\frac{G(\mathrm{Q}^{2-}) - G(\mathrm{Q})}{2F} - E^{\mathrm{abs}}_{\mathrm{SHE}} \tag{13}$$

constant	value
G*(H⁺, aq), 1 M	−270.3 kcal/mol
E_{SHE} (absolute)	4.28 V, same convention
RT ln 10	1.364 kcal/mol
F	23.06 kcal/(mol·V)

Reference values. G^*(H⁺) and the absolute SHE potential must come from the same convention (previous slide). They shift every absolute pK_a or potential by a constant, so differences between molecules are unaffected. With consistent values, the proton and SHE terms of (12) combine into the experimental G of H_2(g).



Thermodynamic consistency: a bookkeeping check

Subtracting (12) from (13) and using (11) for both deprotonations of QH_2 gives an identity between the three observables:

$$E^{\circ\circ} = E^{\circ} - \frac{RT\ln 10}{2F}\left(\mathrm{p}K_{\mathrm{a}1} + \mathrm{p}K_{\mathrm{a}2}\right), \qquad \frac{RT\ln 10}{2F} = 0.0296\ \mathrm{V} \quad (298.15\ \mathrm{K}) \tag{14}$$

(14) follows algebraically from (11) to (13), so it holds to rounding error whenever all terms come from one table of G values. A non-zero residual means an arithmetic or unit error, or different constants in different formulas.

It says nothing about the accuracy of the method, and it does not detect a state that was never calculated. The real test is comparison with experiment.

Different methods. Combining pK_a values from one method with potentials from another breaks (14) by roughly the methods' disagreement, so the residual measures their consistency.

Example for benzoquinone: from the experimental E° and pK_a values, $E^{\circ\circ}$ = 0.70 V − 0.0296 V × (10.2 + 11.7) = 0.05 V, the plateau of the E(pH) line in Part 0.

Accuracy, and how an error reaches the observables

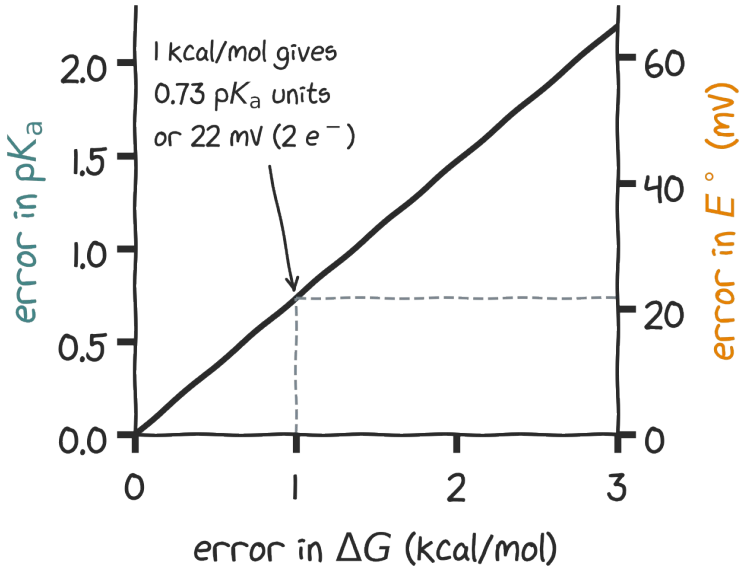
Every observable in this series is a Gibbs energy difference. An error $\delta\Delta G$ propagates as

$$\mathrm{p}K_\mathrm{a} = \frac{\Delta G_\mathrm{deprot}}{RT\ln 10} \quad\Rightarrow\quad \delta\mathrm{p}K_\mathrm{a} = \frac{\delta\Delta G}{RT\ln 10} = 0.733\,\frac{\delta\Delta G}{\mathrm{kcal\,mol^{-1}}} \tag{15}$$

$$E = -\frac{\Delta G_\mathrm{red}}{zF} \quad\Rightarrow\quad |\delta E| = \frac{|\delta\Delta G|}{zF} = \frac{43.4\ \mathrm{mV}}{z}\,\frac{\delta\Delta G}{\mathrm{kcal\,mol^{-1}}} \tag{16}$$

Hybrid functionals, organic molecules: bond lengths to about 0.01 to 0.02 Å; harmonic frequencies a few per cent above measured fundamentals, partly because real vibrations are anharmonic (scale factors about 0.96 for fundamentals, about 0.98 for the ZPE); reaction energies to a few kcal/mol.

Largest error for pK_a and potentials: the continuum-solvation Gibbs energy of ions, often 3 to 5 kcal/mol for anions, that is 2 to 4 pK_a units. Further sources: self-interaction error (an electron partly repels itself, so excess charge spreads too far), missing dispersion, and near-degenerate configurations. Errors partly cancel between related molecules, so relative values are more reliable than absolute ones.



J. Ho and M. L. Coote, WIREs Comput. Mol. Sci. 1, 649 (2011); A. V. Marenich et al., Phys. Chem. Chem. Phys. 16, 15068 (2014). Constants: RT ln 10 = 1.364 kcal/mol and F = 23.06 kcal/(mol·V) at 298.15 K.

Division of labour between Epik, Jaguar and the analysis

EPIK 7

Which forms exist?

Moves protons only. Seconds or less per molecule.

Only its list of states is used

JAGUAR

What is each form's G?

DFT in water with frequencies. Hours per state.

Most of the computer time

ANALYSIS (S3 BY HAND, S4 SCRIPT)

pK_a and potentials

pK_a, E°, E^{∞}, populations, E(pH) and the consistency check, computed in milliseconds.

Arithmetic on the G table

Epik only has to avoid missing a state. An Epik pK_a error does not enter the computed values; it matters only if it hides a state, which is why Epik is run wide and its list inspected.

Part A · Hands-on in Maestro

Project, structures, LigPrep, and a first look at the Jaguar panel

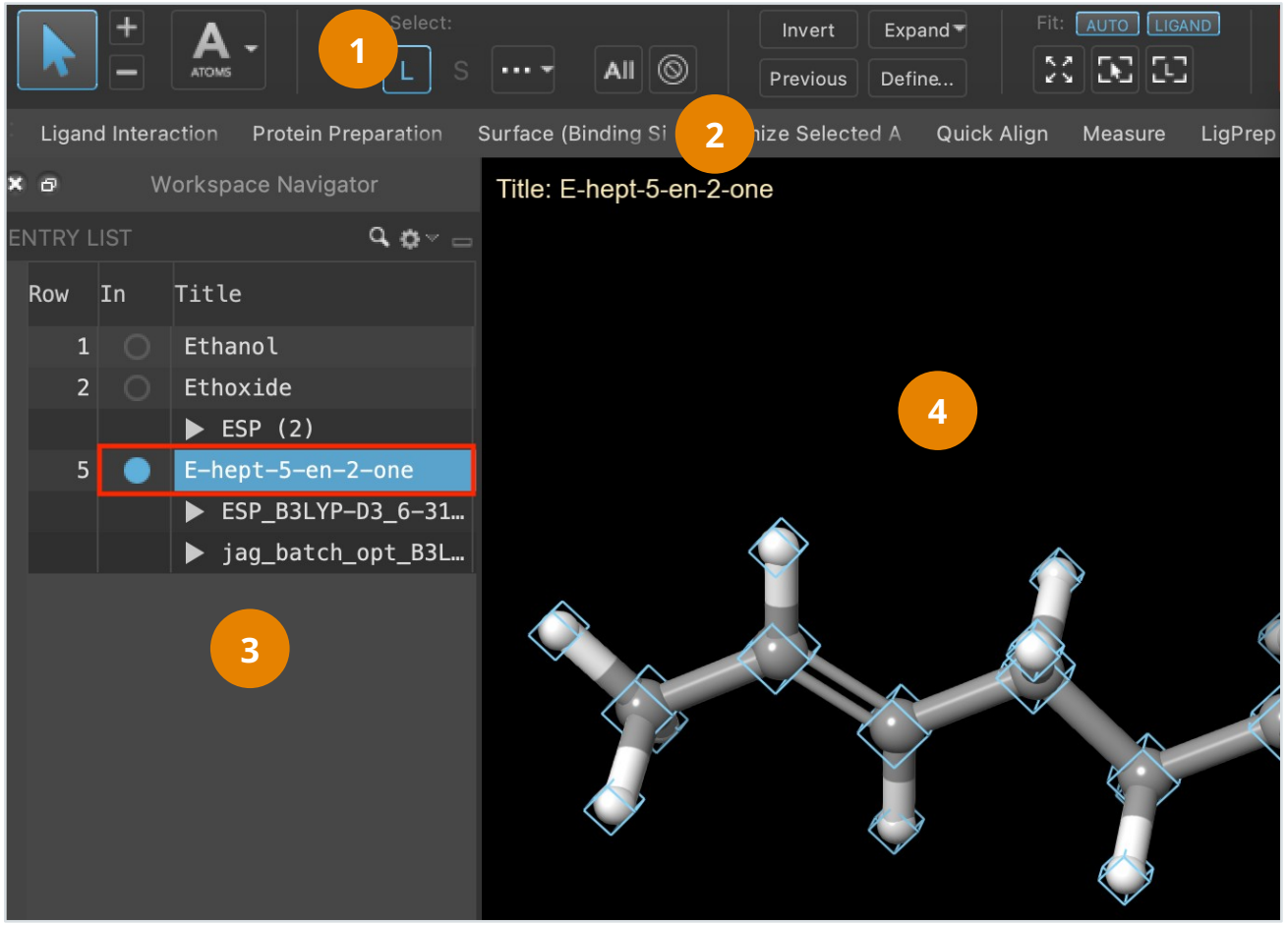
Schrödinger modules used in this series

tool	its job in our workflow	sessions
Maestro	the interface: build, look, launch jobs, inspect results	1 - 5
LigPrep	SMILES to a clean 3D structure, ionisation state kept exactly as written	1, 4
Epik 7	lists protonation states and tautomers	1, 5
Jaguar	DFT: geometry optimisation and frequencies in water, giving G	2 - 5
MacroModel	conformer search; for these small molecules OH rotamers are checked by hand	not used
Job Control (jsc)	sends jobs to the compute hosts and tracks them	1 - 5
run + Python API	your own scripts, run with the suite's Python	4, 5

Materials Science is the Maestro profile we work in, not a separate program. Suite version everywhere: 2026-2.



The Maestro main window

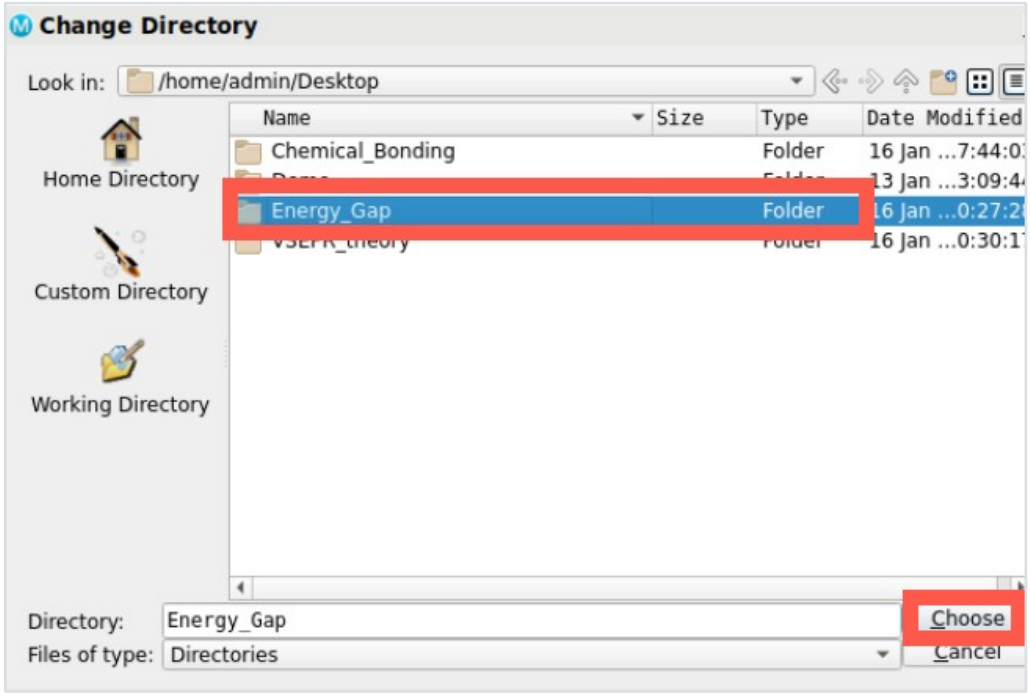


1. **Toolbar:** selection, styles, fit view, build tools
2. **Favourites bar:** one-click panels, including LigPrep
3. **Entry List:** every structure in the project; the dot puts it in the Workspace
4. **Workspace:** what is shown here is what a panel takes as input

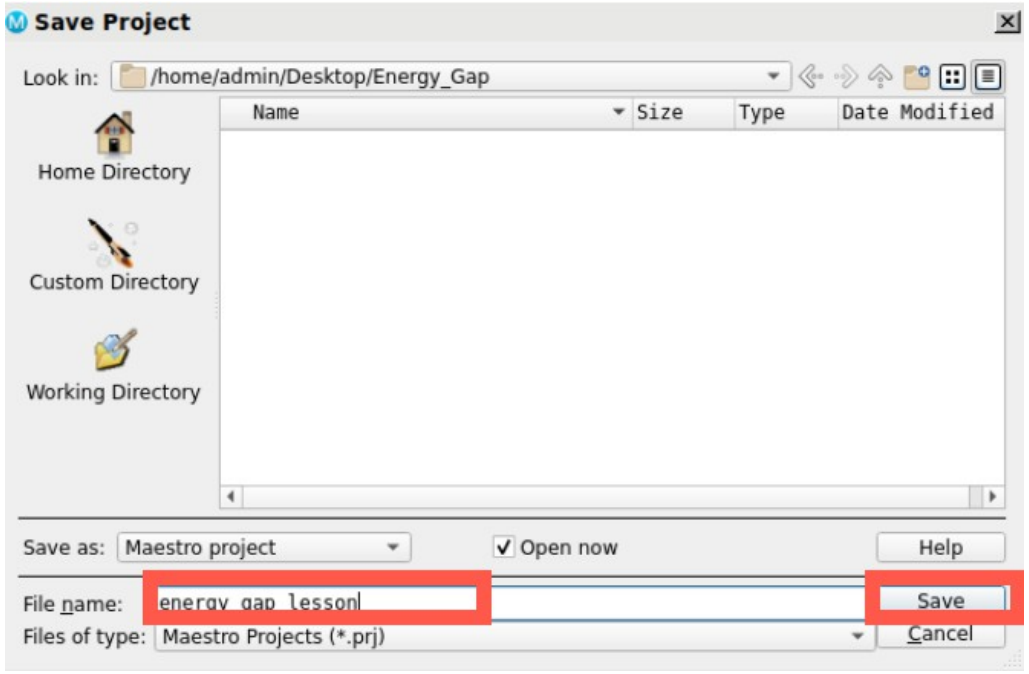
- **Task search** (top right, A2) opens any panel by name
- **Job Monitor** (status bar) shows running jobs and output files

Screenshot: Schrödinger, Teaching with Schrödinger lesson plan (2026). Red boxes are from the original lesson.

A1 · Setting up a clean project



`File > Change Directory`

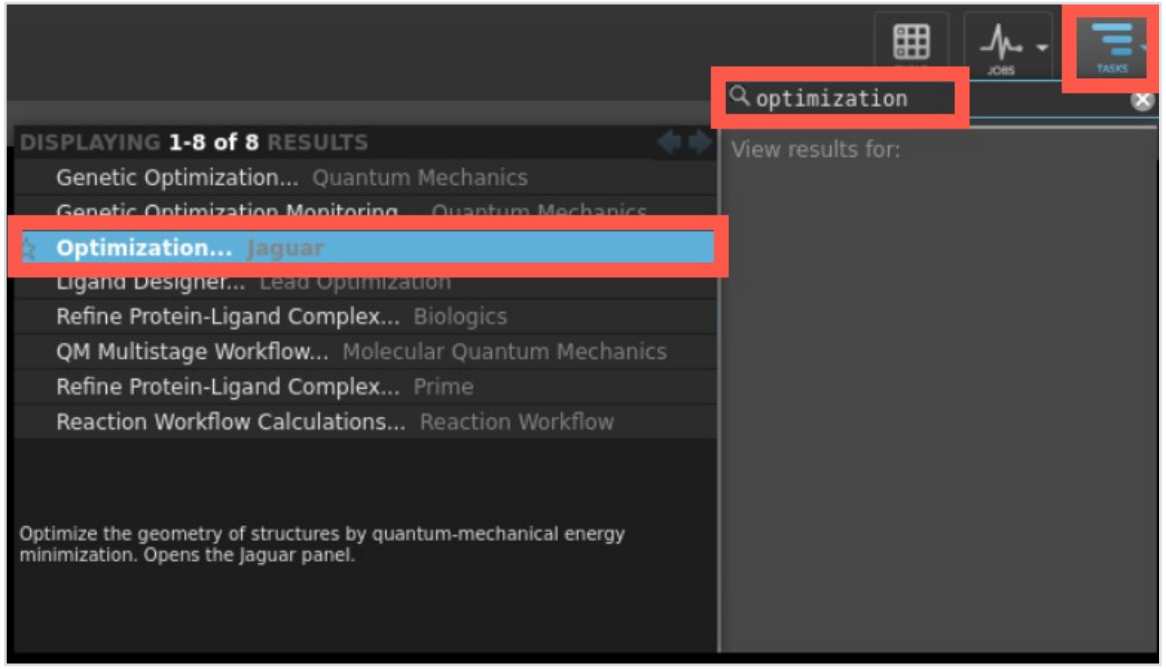


`File > Save Project As`

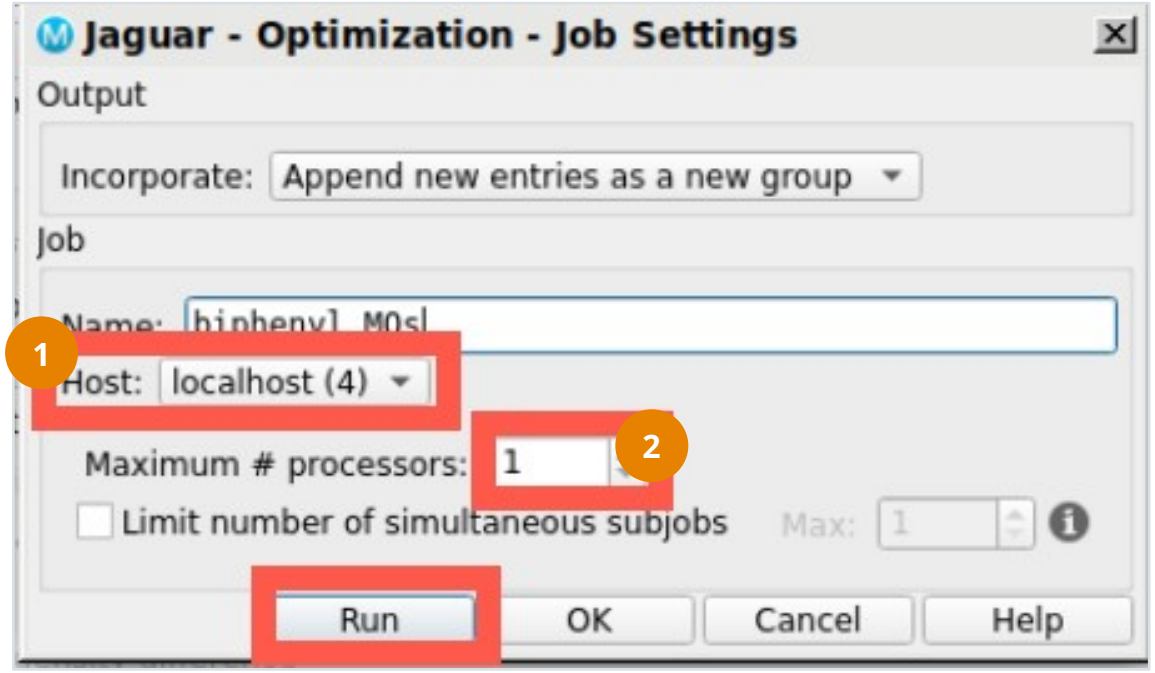
1. Download the ZIP, extract it to C:\ and rename the folder to schrodinger-tutorials-redox
2. Create 01-introduction\results, then Change Directory to 01-introduction
3. Save Project As bq_part1
4. Help > About Maestro: it must say 2026-2

Repository: github.com/NabKh/schrodinger-tutorials-redox (Code > Download ZIP). Work on the local disk: job control is unreliable on network shares. Screenshots: Schrödinger, Teaching with Schrödinger (2026).

A2 · Task search and job settings



Task search: type Epik, LigPrep or optimization. Faster than the menus.



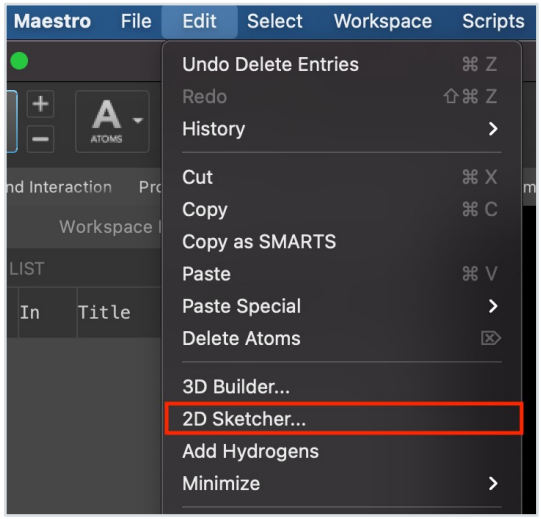
1 Host: localhost for LigPrep and Epik; a compute host for Jaguar.

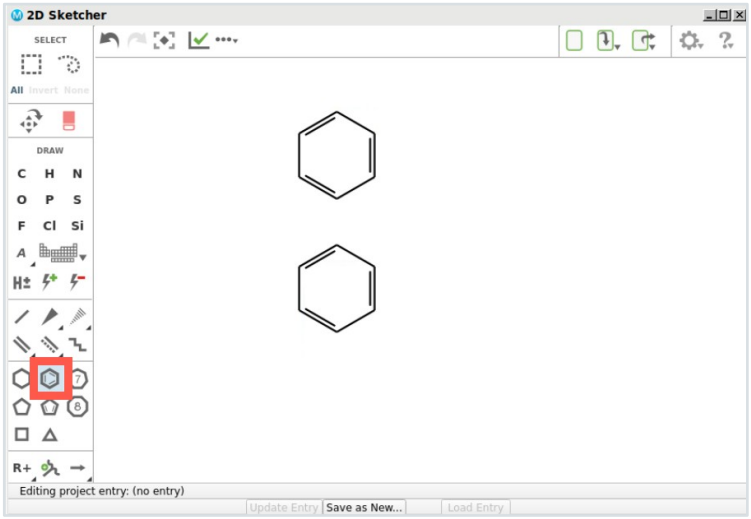
2 Processors: licence tokens are shared; in a group session limit the number of simultaneous jobs.

Job name: bq_<state>_<what>, so files are findable.

Screenshots: Schrödinger, Teaching with Schrödinger lesson plan (2026). The lesson runs everything on localhost; DFT jobs here go to a compute host.

A3 · Drawing versus importing structures





Way 1, draw it: Edit > 2D Sketcher, draw the ring and the oxygens, Save as New. Good for one molecule; easy to get a charge or a hydrogen wrong.

Way 2, import SMILES (what we use)

```
O=C1C=CC(=O)C=C1   bq_O
Oc1ccc(O)cc1       bq_QH2
```

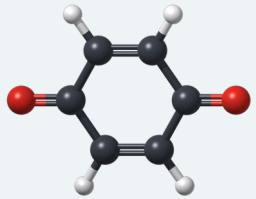
Every state is written exactly, charge included. File > Import Structures reads the .smi file and creates one entry per line.

Scripts in Session 4 start from the same file, so GUI and scripts use identical inputs.

Screenshots: Schrödinger, Teaching with Schrödinger lesson plans (2026).

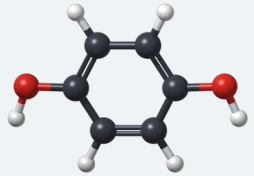
A3 · Importing the benzoquinone states

```
# inputs\bq_closed_shell.smi
O=C1C=CC(=O)C=C1      bq_Q
Oc1ccc(O)cc1          bq_QH2
[O-]c1ccc(O)cc1       bq_QHm
[O-]c1ccc([O-])cc1    bq_Q2m
# inputs\bq_radicals.smi
[O]c1ccc(O)cc1        bq_QHrad
[O-]c1ccc([O])cc1     bq_Qradm
```



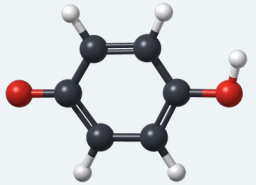
`bq_Q`

Q, charge 0



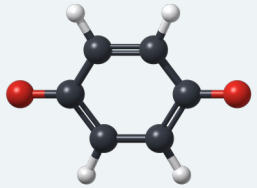
`bq_QH2`

QH_2, charge 0



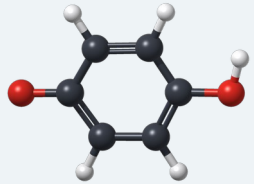
`bq_QHm`

QH^-, charge −1



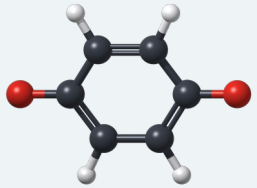
`bq_Q2m`

Q^{2-}, charge −2



`bq_QHrad`

QH•, 0, doublet



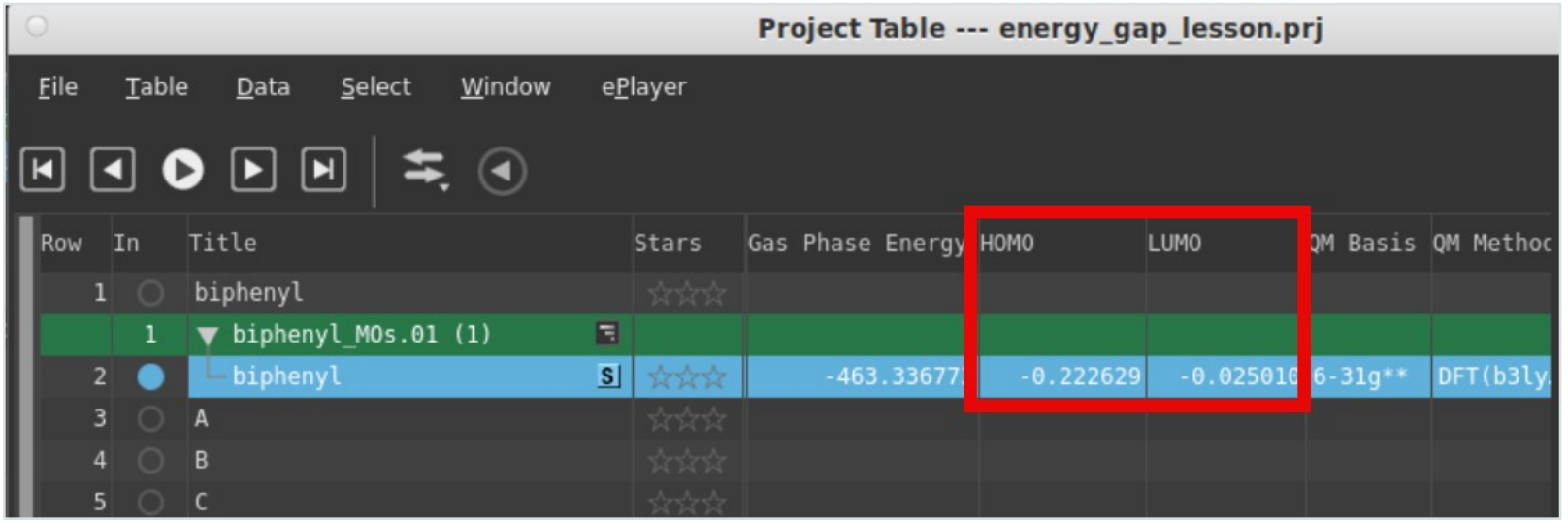
`bq_Qradm`

$Q^{\bullet -}$, −1, doublet

1. Open the two files in inputs\ with a text editor
2. File > Import Structures, both files: six entries appear
3. Select the two radicals; File > Export Structures > results\ bq_radicals.mae
4. Which entries are radicals, and how can you tell from the SMILES?

The charge is in the structure; the spin is set in Jaguar.

A4 · Checking the entries in the Project Table



Title

bq_Q, bq_QH2 … exactly the key names

Formula

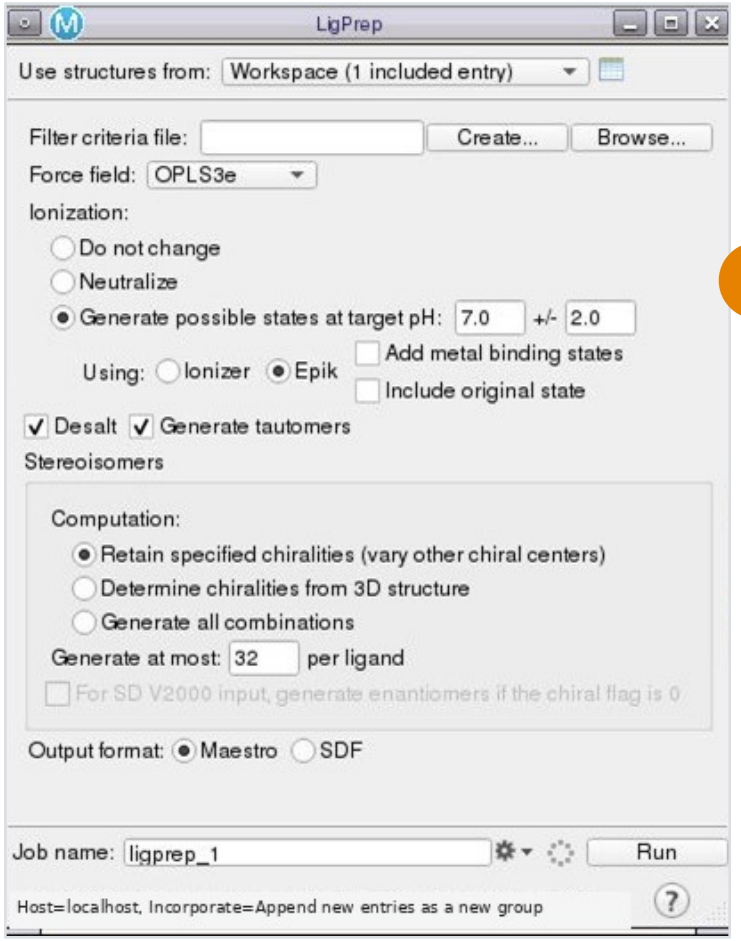
C6H4O2 for Q, C6H6O2 for QH_2, one H less per deprotonation

Total charge

Q 0, QH_2 0, QH^- −1, Q^{2-} −2, QH• 0, $Q^{\bullet -}$ −1

Screenshot: Schrödinger, Teaching with Schrödinger (2026), with Jaguar results as columns. If formula and charge columns are not shown, scripts/check_structures.py gives the same check (Part B).

A5 · LigPrep with the ionisation state kept



2

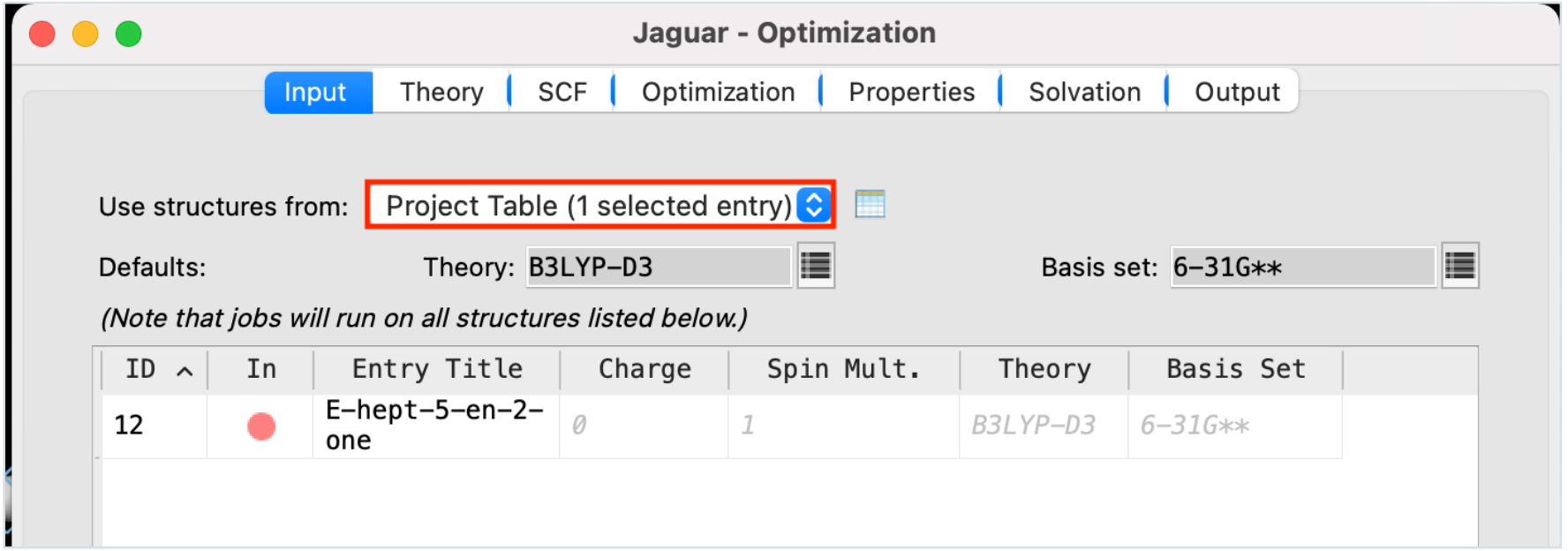
In the LigPrep panel

1. **Use structures from:** the four closed-shell entries, not the radicals
2. **Ionization:** Do not change: the SMILES already defines the state
3. **Tautomers:** untick Generate tautomers
4. **Stereoisomers:** 1 per ligand (benzoquinone has none)
5. **Job:** localhost, name bq_prep, then Run

Caution: with target pH 7, LigPrep turns QH^- back into neutral hydroquinone without a warning. Radicals are left out because LigPrep adds hydrogens to them.

Screenshot: Texas A&M HPRC, Drug Docking with Schrödinger (2021), an older release; check the option names in yours.

A6 · The Jaguar Optimization panel (preview of Session 2)



Input

which entries; charge and spin per entry

Theory

B3LYP, def2-TZVPD; spin-unrestricted for radicals

Properties

vibrational frequencies on

Solvation

PBF, water

Job settings

compute host, processors

*Screenshot: Schrödinger, Teaching with Schrödinger lesson plan (2026). The defaults shown (B3LYP-D3, 6-31G**) are not the level used in this tutorial.*

Part B · The command line

The same steps as commands that can be saved, rerun and checked

Opening the 2026-2 suite from PowerShell

WINDOWS WORKSTATION · POWERSHELL

```
$S = "C:\Program Files\Schrodinger2026-2"
cd C:\schrodinger-tutorials-redox\01-introduction
& "$S\run" python3 -c "print('suite OK')"
suite OK
```

LINUX

```
export SCHRODINGER=/opt/schrodinger2026-2
$SCHRODINGER/run python3 -c "print('suite OK')"
```

Green lines are what the computer prints back.

Do not rely on $env:SCHRODINGER

If more than one release is installed, it may point at another one. Results from different releases should not be mixed. Always use the full 2026-2 path, stored in $S.



Compute hosts, job server and job control

```
# hosts you can submit to
& "$S\jsc" hosts list
# once per user: register with the job server
# (server and port from your administrator)
& "$S\jsc" cert get <server>:<port>
# your running and finished jobs
& "$S\jsc" list
```

host	for Jaguar?	why
<compute host>	yes	CPU queue with suite 2026-2
<second host>	yes	same suite, use when the first is busy
localhost	no	your workstation; fine for LigPrep and Epik
GPU queues	no	not needed for DFT on these molecules

Licence tokens and throughput

Licence tokens are shared by everyone on the same licence server, so a job can wait for a token even when cores are free. In a group session, limit the number of simultaneous jobs and check jsc list before submitting more.



LigPrep from the command line

```
& "$S\ligprep" -ismi inputs\bq_closed_shell.smi -omae results\bq_prepped.maegz `
      -i 0 -nt -s 1 -NJOBS 1 -WAIT
# formula, charge, electron count and multiplicity of every entry
& "$S\run" scripts\check_structures.py results\bq_prepped.maegz results\bq_radicals.mae
```

flag	what it does
`-ismi / -omae`	input SMILES file, output Maestro file
`-i 0`	keep the ionisation state; otherwise LigPrep neutralises (and may re-ionise) QH^- and Q^{2-}
`-nt`	no tautomers: one output per input
`-s 1`	one stereoisomer per input
`-NJOBS 1 -WAIT`	one subjob; return only when the job has finished
`` ` ``	PowerShell line continuation (key left of 1): one command over two lines

Expected: 56 electrons for bq_Q, 57 for bq_QHrad and bq_Qradm (multiplicity 2), 58 for bq_QH2, bq_QHm and bq_Q2m.



GUI actions and their command-line equivalents

in Maestro	on the command line
File > Change Directory	`cd <folder>`
File > Export Structures (format only)	`& "$S\utilities\structconvert" in.maegz out.mae`
LigPrep panel	`& "$S\ligprep" -ismi ... -omae ... -i 0`
Epik panel	`& "$S\epikx" in.mae out.maegz -ph 7.0 -pht 7.0`
Jaguar panel > Run	`& "$S\jaguar" run -HOST <host> -PARALLEL 16 job.in`
Job Monitor	`& "$S\jsc" list`

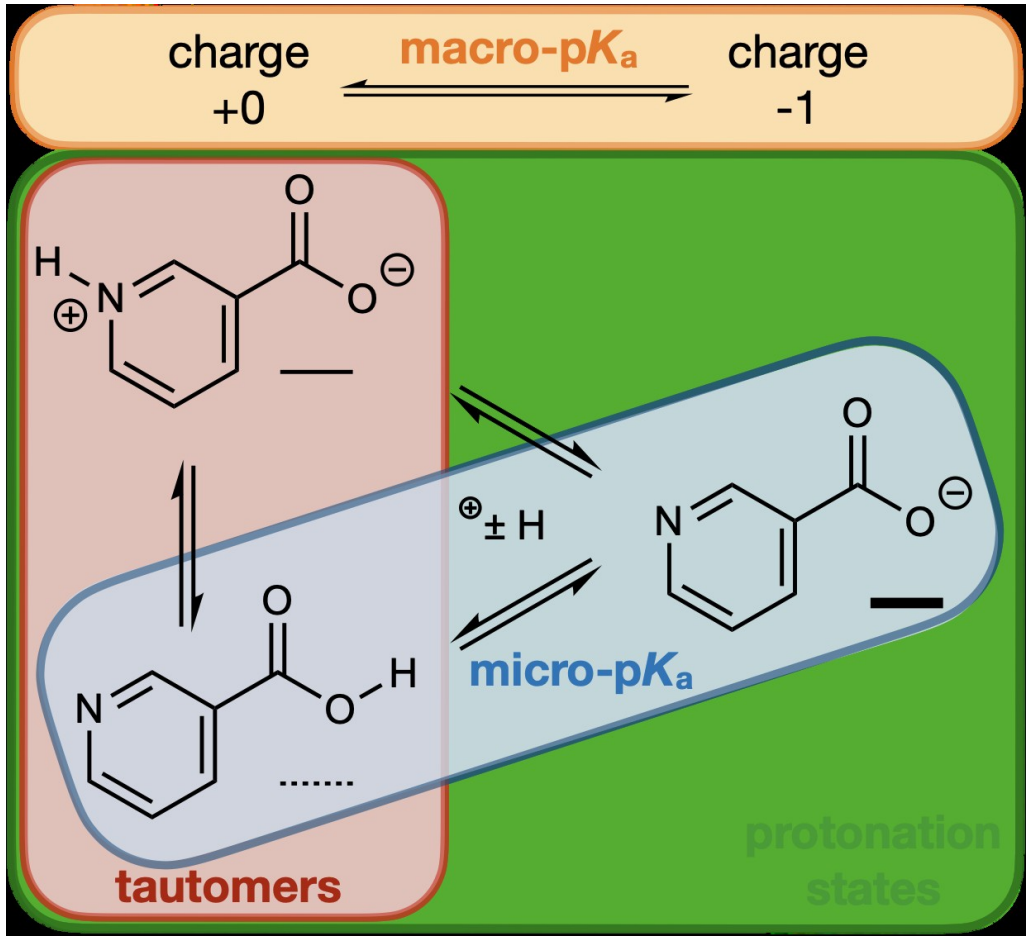
The engines and files are identical. The GUI writes the input file and calls these programs; a script calls the same programs directly.



Part C · Epik

What it lists and where it stops

Micro-pK_a, macro-pK_a and tautomers



Micro-pK_a

between two specific structures that differ by one proton

Tautomers

same formula and charge, the proton sits elsewhere

Macro-pK_a

between charge states, summing over every structure of each charge

$$\mathrm{p}K_\mathrm{a}^\mathrm{macro} = -\log_{10} \sum_i 10^{-\mathrm{p}K_{\mathrm{a},i}}$$

One acid, two equivalent sites at 9.0: $9.0 - \log_{10} 2 = 8.70$. With several acid microstates, use the general form (appendix).

Figure: Schrödinger white paper, Small molecule protonation state enumeration and pK$_a$ prediction (2023).

Epik 7 and its default settings

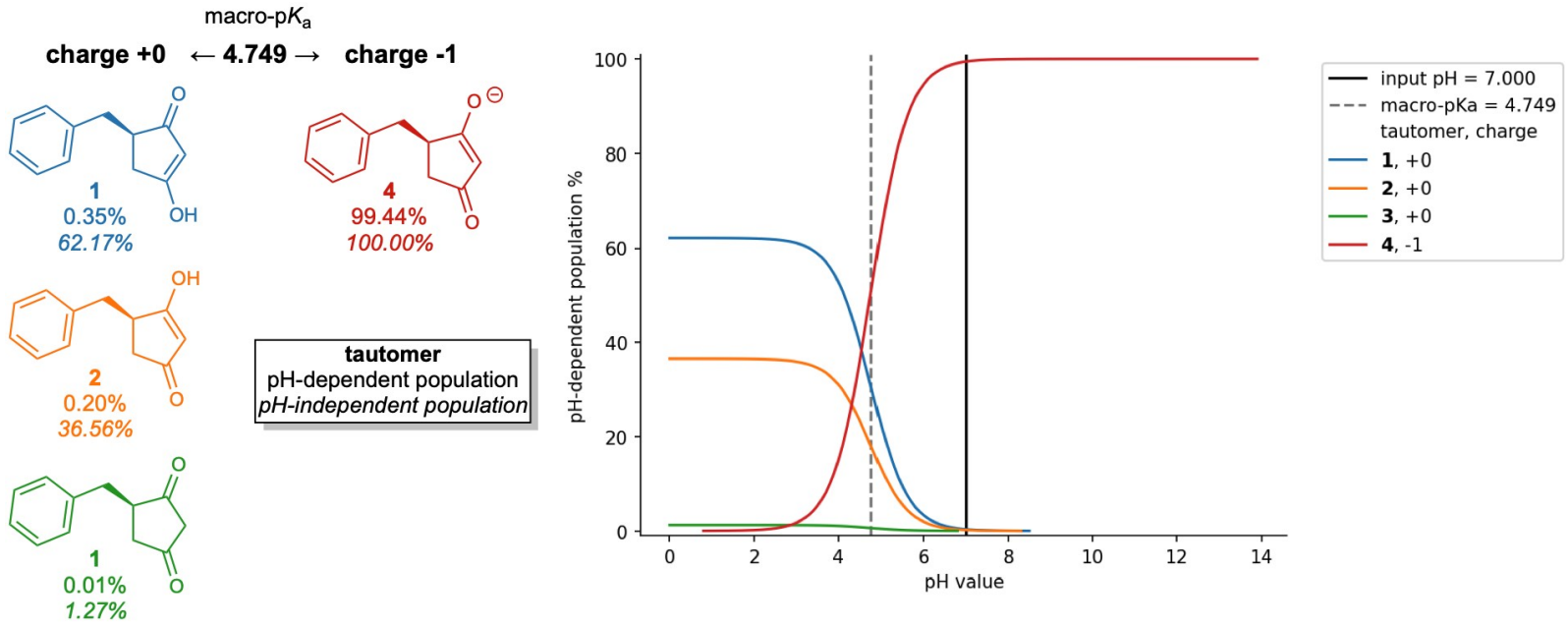
- **Machine learning.** An ensemble of graph neural networks predicting pK_a site by site (Johnston et al., JCTC 2023).
- **Trained on over 42,000 pK_a values,** experimental and computed with Jaguar pK_a, mostly for drug-like, closed-shell molecules.
- **Fast:** seconds or less per molecule.
- **Protons only.** It never changes the number of electrons.
- **Output:** protonation states and tautomers, each with a population at the chosen pH.

Default settings and the wide run

setting	default (2026-2)	wide run
target pH	7.4	7.0
pH tolerance	**2.0**	**7.0**
population cut-off	alternative to the tolerance	not set
charge window	−2 to +2	QH_2 (0) to Q^{2-} (−2)
max. structures	16	16

Epik's job here is the list of states. Each listed state gets a Jaguar Gibbs energy, so Epik only has to avoid missing one: that is why it is run wide.

Epik output: populations against pH



Each structure

gets a population at the input pH and across pH 0 to 14

The dashed line

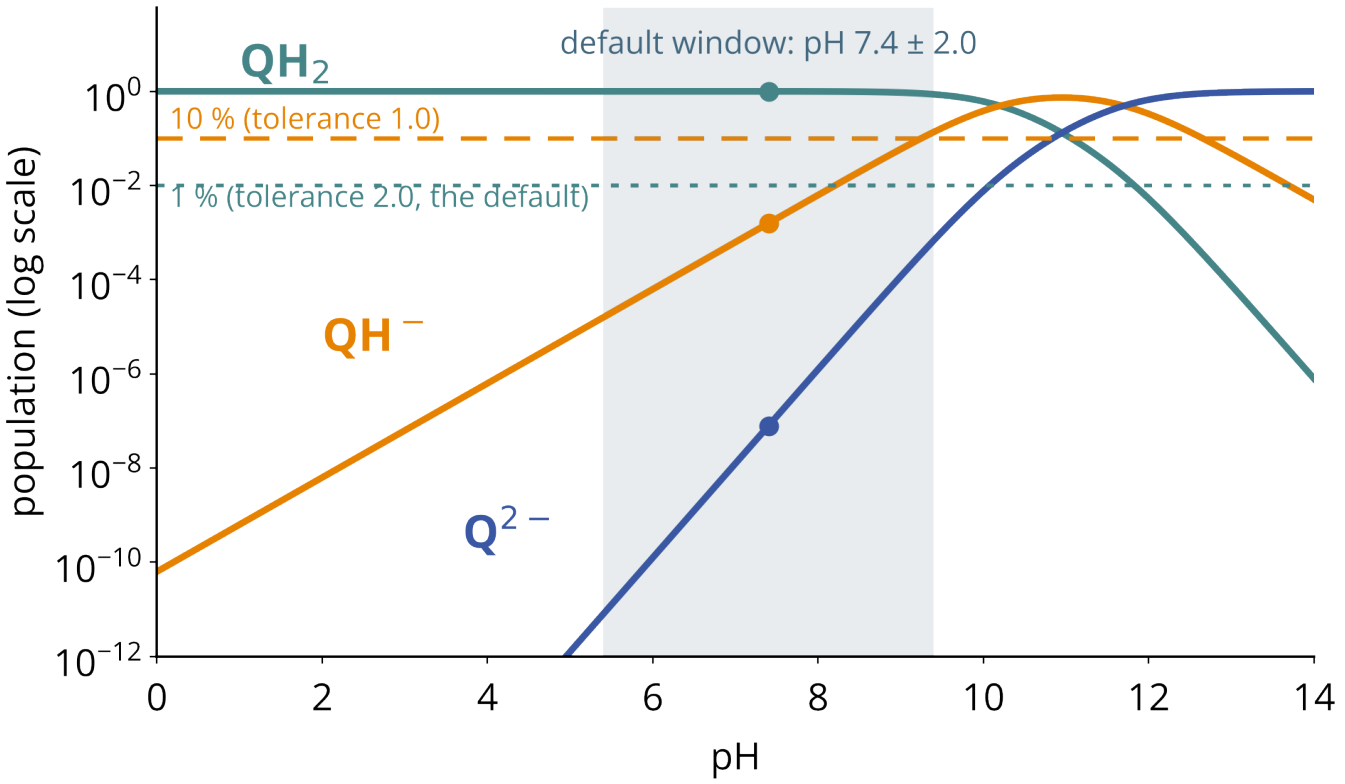
is the macro-pK_a between the two charge states

Tautomers

1, 2 and 3 share a charge; their split does not depend on pH

Figure: Schrödinger white paper (2023), Epik 7 output example.

Populations of reduced benzoquinone and the default window



Populations from the experimental pK_a 10.2 and 11.7; Epik uses its own predicted pK_a values.

99.8 %

QH_2 at pH 7.4

0.16 %

QH^- at pH 7.4

0.000008 %

Q^{2-} at pH 7.4

Threshold at pH 7.4 (Epik 7 paper): only QH_2 passes. Populated anywhere in 5.4 to 9.4: QH^- also passes, 14 % at pH 9.4. Run 1 shows which applies.

C1 · Three Epik runs on QH_2 in Maestro

1. Select bq_QH2 in the group made by the bq_prep job; open Epik through task search, input: selected entries
2. Run 1: all defaults. Job name bq_QH2_epik_default
3. Run 2: target pH 7, pH tolerance 7, up to 16 states. Job name bq_QH2_epik_wide
4. Run 3: target pH 12, pH tolerance 2. Job name bq_QH2_epik_ph12
5. Count the output entries and read their total charge

Your prediction and the result

	states predicted	states observed	charges observed
Run 1 default			
Run 2 wide			
Run 3 pH 12			

What to look for: threshold reading: Run 2 adds QH^-, Run 3 gives QH^- and Q^{2-}. Window reading: Runs 2 and 3 give all three. No run gives Q or QH• (three slides on).

C2 · The same three runs from the command line

```
# take bq_QH2 out of the LigPrep output, then check the options of this release
& "$S\run" scripts\extract_entry.py results\bq_prepped.maegz bq_QH2 results\bq_QH2.mae
& "$S\epikx" -h
# Run 1, defaults
& "$S\epikx" results\bq_QH2.mae results\bq_QH2_epik_default.maegz -WAIT
# Run 2, wide window
& "$S\epikx" results\bq_QH2.mae results\bq_QH2_epik_wide.maegz -ph 7.0 -pht 7.0 -ms 16 -WAIT
# Run 3, window centred at pH 12
& "$S\epikx" results\bq_QH2.mae results\bq_QH2_epik_ph12.maegz -ph 12.0 -pht 2.0 -ms 16 -WAIT
```

flag	meaning
`-ph`	target pH
`-pht`	pH tolerance: states within pH ± tolerance
`-ms`	maximum number of returned structures (default 16)

Check on your release

Expected flags; confirm them with -h on your release. The pH tolerance and a population cut-off are alternatives: set one, not both.



Enumeration in Schrödinger, step by step

1. **Build one input per redox level,** n_e = 0, 1, 2, with charge and hydrogens as intended. Build the open-shell row (QH•, Q•⁻) by hand: Epik and LigPrep are made for closed-shell molecules.

2. **Run Epik on each closed-shell input:** pH 7 ± 7, plus pH 12 ± 2 and pH 2 ± 2: under the threshold reading, pH 7 ± 7 misses states populated only near the ends. Tautomers are on by default.

3. **Or LigPrep with Epik:** Ionization with Epik (not Epik Classic), tautomers on; confirm the command-line flags with ligprep -h. The opposite of step A5.

4. **Command line:** epikx with -ph, -pht and -ms; confirm the flags with -h first.

5. **Collect and check:** group by charge, remove duplicates and symmetry-equivalent copies (statistical factor through σ or by counting, not both), check formula and charge, set the multiplicity, check OH rotamers. Each microstate gets a Jaguar job.

Always inspect the list before submitting Jaguar jobs: it depends on the pK_a values Epik predicts and on the window.

What to expect

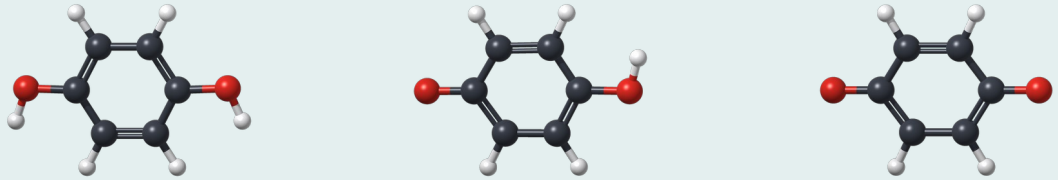
Benzoquinone over pH 0 to 14: Q (n_e = 0; QH^+ only far below pH 0), QH• and $Q^{\bullet -}$ (n_e = 1, built by hand), and QH_2, QH^- and Q^{2-} (n_e = 2), Q^{2-} possibly only from the pH 12 run.

2-Methylhydroquinone: the n_e = 2 row gives two monoanion microstates; both need a Gibbs energy.

Methyl viologen: each row gives its input only.

Why Epik cannot cross the redox axis

Epik can reach: same electrons, different protons



QH_2 QH^- Q^{2-}

Epik cannot reach from QH_2: different electron count



Q QH•

different rows of the proton-electron grid

One structure per redox level, then Epik on each

```
for n_e in (0, 1, 2):
    s = structure_at(n_e)
    if n_e % 2 == 0:   # closed shell
        states += epik(s)
    else:              # radicals by hand
        states += [s]
```

Running Epik once on the neutral parent gives only one row of the proton-electron grid. In Session 5 you script this loop.

Good practice

1. **Run LigPrep and Epik locally, Jaguar on a compute host.** localhost is the workstation, not a compute node.
2. **Work in a local folder.** Job control is unreliable on network shares.
3. **Call the suite by its full 2026-2 path.** $env:SCHRODINGER may point at another release.
4. **Keep the ionisation state in LigPrep.** 'Do not change' or -i 0, tautomers off (-nt), radicals left out; then check formula and charge.
5. **Run Epik wide when you need a list.** pH 7 ± 7 plus runs near pH 2 and 12, then inspect every state.



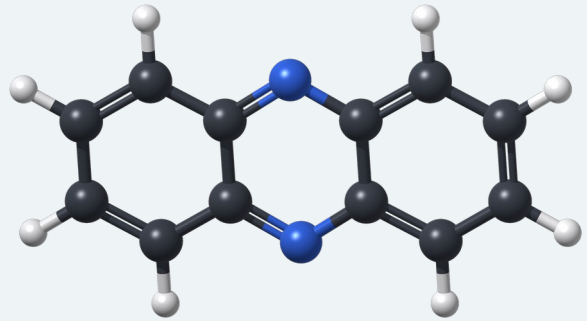
Homework before Session 2

1. **Work through** session1-notebook.ipynb in 01-introduction: populations, E(pH) line, macro-pK_a, error propagation.

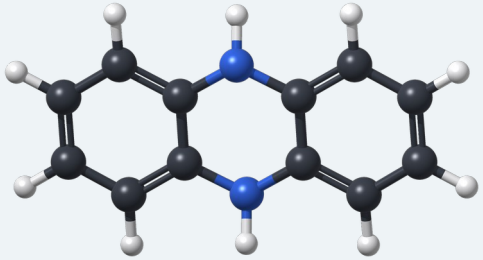
2. **Draw the proton-electron grid** of phenazine: which cells are populated between pH 0 and 14? Give SMILES, n_H, n_e, charge and multiplicity of each.

3. **Run Epik** with default and wide settings on dihydrophenazine, in the GUI or typed. Bring state counts and charges.

4. **Five lines** on what surprised you and what is unclear. We start Session 2 with them.



phenazine `c1ccc2nc3ccccc3nc2c1`



dihydrophenazine `c1ccc2c(c1)Nc1ccccc1N2`

Preview of Session 2: the Jaguar input file

bq_QH2.in (the &gen section)

```
&gen
dftname=B3LYP
basis=def2-TZVPD
isolv=2
solvent=water
nogas=2
igeopt=1
ifreq=1
molchg=0
multip=1
&
```

keyword	meaning
`dftname, basis`	level of theory used in this tutorial
`isolv, solvent`	PBF implicit water
`nogas=2`	no gas-phase optimisation
`igeopt, ifreq`	optimise, then frequencies, giving G
`molchg, multip`	the two rules from Part 0

In Session 2: Q and QH_2 at PBE/LACVP**++ (6-31++G**, spherical d; fast), B3LYP and B3LYP-D3/def2-TZVPD. Radicals: multip=2, iuhf=1. For the IR comparison, the same input without isolv, solvent and nogas.



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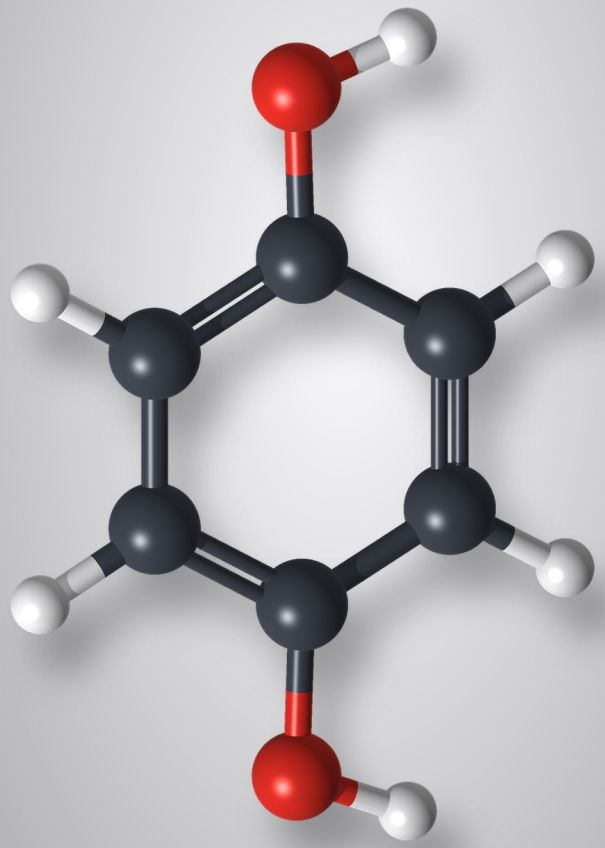
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Questions and discussion

Next: Session 2 · Geometry optimisation and harmonic frequencies in continuum solvent

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DIFFER

Appendix · Formal details

Definitions and equations referred to in Part 1

Appendix · Constrained search and the SCF equations

Universal functional (Levy): the minimum over all antisymmetric Ψ that give the density *n*:

$$F_{\mathrm{HK}}[n] = \min_{\Psi \to n} \langle \Psi | \hat{T}_e + \hat{V}_{ee} | \Psi \rangle \tag{A1}$$

Matrix form of the Kohn-Sham equations (as in Roothaan-Hall), with the density matrix **P** and occupations f_i:

$$\mathbf{F}(\mathbf{P})\,\mathbf{C} = \mathbf{S}\,\mathbf{C}\,\boldsymbol{\varepsilon}, \qquad F_{\mu\nu} = \langle \chi_\mu | \hat{f}_{\mathrm{KS}} | \chi_\nu \rangle, \quad S_{\mu\nu} = \langle \chi_\mu | \chi_\nu \rangle, \quad P_{\mu\nu} = \sum_{i}^{\mathrm{occ}} f_i \, C_{\mu i} C_{\nu i}^{*} \tag{A2}$$

Self-consistent field: iterate until the energy (and the density) change by less than a threshold:

$$\mathbf{P}^{(k)} \longrightarrow \mathbf{F}\big(\mathbf{P}^{(k)}\big) \longrightarrow \mathbf{C}^{(k+1)}, \boldsymbol{\varepsilon}^{(k+1)} \longrightarrow \mathbf{P}^{(k+1)}, \qquad \text{until } |E^{(k+1)} - E^{(k)}| < \tau_E \tag{A3}$$



M. Levy, Proc. Natl. Acad. Sci. USA 76, 6062 (1979); C. C. J. Roothaan, Rev. Mod. Phys. 23, 69 (1951); G. G. Hall, Proc. R. Soc. A 205, 541 (1951).

Appendix · Exchange, dispersion, general macro-pK_a

Exact (Hartree-Fock) exchange from the occupied orbitals, summed over pairs of equal spin:

$$E_{\mathrm{x}}^{\mathrm{HF}} = -\frac{1}{2} \sum_{i,j}^{\sigma_i=\sigma_j} \iint \frac{\varphi_i^*(\mathbf{r})\varphi_j(\mathbf{r})\,\varphi_j^*(\mathbf{r}')\varphi_i(\mathbf{r}')}{|\mathbf{r}-\mathbf{r}'|}\, d\mathbf{r}\, d\mathbf{r}' \tag{A4}$$

D3 dispersion correction, pairwise with damping at short range:

$$E_{\mathrm{disp}}^{\mathrm{D3}} = -\sum_{A<B} \sum_{n=6,8} s_n \frac{C_n^{AB}}{R_{AB}^n} f_{\mathrm{damp},n}(R_{AB}) \tag{A5}$$

Macroscopic pK_a for any number of acid microstates HA_i and base microstates A_j in one charge step:

$$\mathrm{p}K_{\mathrm{a}}^{\mathrm{macro}} = \frac{1}{RT\ln 10}\left[G(\mathrm{H}^+) - RT\ln\sum_j e^{-G(\mathrm{A}_j)/RT} + RT\ln\sum_i e^{-G(\mathrm{HA}_i)/RT}\right] \tag{A6}$$



S. Grimme et al., J. Chem. Phys. 132, 154104 (2010).